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Contents

1	Light and Gaussian Modes	5
1.1	Plane Waves	5
1.2	Gaussian Modes	6
1.3	The Paraxial Wave Equation	7
1.4	The fundamental Gaussian mode	9
1.4.1	Astigmatism	12
1.5	Gaussian Beams Traversing optical Layouts	13
1.6	Higher Order Modes and optical imperfections	14
1.7	Concluding remarks	19
2	Balanced Homodyne for LIGO A+	20
2.1	BHD versus DC readout for A+	20
2.2	Local Oscillator Path Length Stability Requirement and HRTS re- quirement	23
2.2.1	Local Oscillator Path	23
2.2.2	Simulation of the Linear Response Function Between a Differ- ential Strain and the Power at the AS Port	26

2.2.3	Converting the Sensitivity Curve for A+ into the HRTS requirement	28
2.2.4	HRTS	30
2.3	Alignment of the LO and AS beam	33
2.4	HAM6 layout	33
2.4.1	The Constraints and Requirements of the Layout of HAM6 . .	33
2.4.2	Chosen Layout	36
2.4.3	WS visualisations as a tool for understanding mode matching problems	38
2.5	Conclusion about BHD for A+	49
3	The Effect of Radius of Curvature Errors in SR3 and SR2 on the Beam Parameter of the Arm Cavity Mode at the SRM.	50
3.1	Introduction	50
3.1.1	Measurements of the Mode Matching Between the Arm Cavities and the OMC at LLO Between 2018-2019.	55
3.2	Modelling Interferometers With Mode Mismatches Using FINESSE . .	58
3.2.1	Thin and Thick Beamsplitters	61
3.3	Modelling of Michelson Interferometer With Fabry-Perot Arm Cavities and A Signal Recycling Cavity	63
3.3.1	Determining Tunings for DC readout	66
3.4	The Beam Parameter of the Arm Mode at the SRM	71
3.5	Power Transmitted Through SRM for a Mode Mismatch Between the Arm Cavity and the SRC	81
3.5.1	Change of finesse of SRC	85

3.6	Conclusion	86
A	FINESSE Simulations	88
A.1	Mode Matching the Laser to the Arm Cavity	88
A.2	Interferometer Parameters	92
A.2.1	Radii of Curvature of optical components	92
A.2.2	Spacings	93
A.2.3	Nominal Simulation	95
A.3	FINESSE Code	97
A.3.1	Michelson Interferometers with Thin Beamsplitter	97
A.3.2	Michelson Interferometers with Thick Beamsplitter	97
A.3.3	Main Simulation	98
B	Algorithm For Finding a Three Optic Solution with 45° Gouy Phase	
	Separation between two of the Optics	103
B.0.1	Method	104
C	Mode Matching Between 2 Astigmatic Beams	109
C.1	Derivation of Mode Matching TM00 Beams	109
C.2	Effect of Error in the Radius of Curvature of a Mirror on the defocus of a Beam	112

Chapter 1

Light and Gaussian Modes

This chapter will show the maths...

- check normalisation constants
- check the formula for the electric fields are actually right!
- change formulas to be in terms of a mode function and something that is to do with its propogational along z, and amplitude.

1.1 Plane Waves

Light is electromagnetic radiation. From Maxwell's equations it can be shown that the electric field, $\mathbf{E}$, component of electromagnetic radiation satisfies the wave equation

$$\nabla^2 \mathbf{E} = \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \mathbf{E}. \quad (1.1)$$

A monochromatic, coherent, plane wave solution for $\mathbf{E}$ satisfies equation (1.1):

$$\mathbf{E} = \hat{\mathbf{n}}E \exp(i(\omega t - \mathbf{k} \cdot \mathbf{r} + \varphi)), \quad (1.2)$$

where $\hat{\mathbf{u}}$ indicates the polarisation, E is the complex field amplitude, ω is the angular frequency, $\mathbf{k}$ is the wave vector and φ is the absolute phase. The physical amplitude of the electric field oscillation is obtained from the $\Re(E)$. For optical systems is useful to define an optical axis. When the light is travelling along the optical axis, the $\mathbf{k} \cdot \mathbf{r}$ becomes kz . k is known as the wave number and is related to the wavelength of the light, λ by

$$k = \frac{2\pi}{\lambda}. \quad (1.3)$$

For plane waves there is no dependence of $\mathbf{E}$ on the x and y coordinates. The absolute phase φ becomes relevant in calculations involving interfering beams. By inserting equation (1.2) into equation (1.1), we obtain the dispersion relationship

$$\omega = ck. \quad (1.4)$$

The plane wave solution can be used to analyse many optical phenomena, such as amplitude modulation, phase modulation and interference.

1.2 Gaussian Modes

Light in an optical set-up is normally generated by a laser. Lasers produce almost monochromatic, coherent, light so in this sense they are quite similar to a plane wave, however lasers also have geometric properties. A laser produces light with an

approximately circular beam profile which is brightest at its centre. The spread of light power in a laser beam follows a Gaussian distribution, and the width of the beam profile is characterised by the radius at which $\sim 86\%$ of the light power is contained.

As the beam propagates along its optical axis, the beam size changes. A Gaussian beam has a point where it is smallest, this is known as the waist of the beam. A Gaussian beam can be completely characterised by the size of its waist and location along the waist along the optical axis. Whenever a Gaussian mode interacts with a simple optical component such as a lens or a mirror, the current Gaussian mode is transformed into a new Gaussian mode.

The following sections describe Gaussian modes and how they interact with real optical layouts with imperfections. The lowest order Gaussian mode will be derived, and the properties of this mode will be explained. The concept of the complex Gaussian beam parameter will be introduced as a way of encapsulating all the information needed to determine the properties of a Gaussian mode. Astigmatic Gaussian beams will be defined in terms of two complex Gaussian beam parameters. The first order Hermite-Gaussian (HG) mode will be shown to model the effect of misalignment, and then it will be described how all laser beams can be made up of a set of HG modes. Finally, the way that a Gaussian mode traverses an optical set-up will be expressed in terms of the ABCD matrix formalism.

1.3 The Paraxial Wave Equation

Short intro on what this is

To derive the paraxial wave equation, we apply the separation of variables to equation (1.1):

$$\mathbf{E}(\mathbf{r}, t) = E(\mathbf{r})T(t). \quad (1.5)$$

Next we reform equation 1.1 in terms of E and T

$$\frac{\nabla^2 E}{E} = \frac{1}{c^2 T} \frac{\partial^2 T}{\partial T^2}. \quad (1.6)$$

By noticing that either side of the equation (1.6) depends on $\mathbf{r}$ or t , the left hand side of this equation must equal a constant, and the Helmholtz equation can be obtained:

$$\frac{\nabla^2 E}{E} = -k^2. \quad (1.7)$$

Rewriting $E(\mathbf{r})$ in terms of two functions, **some how remove units from $u(\mathbf{r})$**)

$$E(\mathbf{r}) = u(\mathbf{r}) \exp(-ikz), \quad (1.8)$$

and substituting this into equation (1.7) yields

$$\left(\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) u(\mathbf{r}) - 2ik \frac{\partial u}{\partial z} + \frac{\partial^2 u}{\partial z^2} \right) \exp(ikz) = 0. \quad (1.9)$$

The paraxial approximation means the $\frac{\partial^2 u}{\partial z^2}$ is neglected as it is much smaller than the other terms. Applying this approximation produces the paraxial Helmholtz equation

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) u(\mathbf{r}) - 2ik \frac{\partial u}{\partial z} = 0. \quad (1.10)$$

The physical interpretation of $u(\mathbf{r})$ is a function that describes the spatial distribution of the complex amplitude of the light field. **reword this so it actually makes sense.**

1.4 The fundamental Gaussian mode

Rewrite E in terms of u .

The simplest geometry of a laser beam can be described by the fundamental Gaussian mode. This is a mode which has a circular intensity profile that follows a Gaussian distribution, and the surface of constant phase is curved. Substituting the simplest solution to (1.10) into equation (1.8) produces the fundamental Gaussian mode **find out what the amplitude constant is**

$$E(\mathbf{r}) = E_0 \sqrt{\frac{2}{\pi}} \frac{1}{w(z)} \exp(i\phi(z)) \exp\left(-ik\frac{x^2 + y^2}{2R_C(z)} - \frac{x^2 + y^2}{w^2(z)}\right) \exp(-ikz). \quad (1.11)$$

This is characterised by a function $w(z)$, which gives the size of the beam at position z the optical axis, by $\phi(z)$, which gives the Gouy Phase of the beam at z along the optical axis, and by $R_c(z)$, which gives radius of curvature of the beam at z . The minimum value of $w(z)$ is known as the beam waist, or w_0 .

It is useful to express $w(z), \phi(z)$ and $R_c(z)$ in terms of the Rayleigh range z_R . The Rayleigh range characterises how fast the beam will spread out. A beam with a short Rayleigh length will expand quicker than a beam with a longer Rayleigh length. For example, in order for the beam to not be significantly clipped by the aperture of the test masses, the Rayleigh length of the Gaussian beam in the arm cavities at LIGO

has to be on the order of kilometres. The size of the waist determines the Rayleigh length, a smaller beam waist corresponds to a shorter Rayleigh length. If the waist of a Gaussian beam at $z = 0$, the position along the z axis by which the radius has increased by a factor of $\sqrt{2}$ is known as the Rayleigh range z_R , and is given by

$$z_R = \frac{\pi w_0^2}{\lambda}. \quad (1.12)$$

The radius of the beam as it propagates along the optical axis can be written as

$$w(z) = w_0 \sqrt{1 + \left(\frac{z}{z_R} \right)^2}. \quad (1.13)$$

The Rayleigh range can be used to approximately split beam into two regions: the near field for $z < z_R$ and the far field for $z > z_R$. In the far field equation (1.13) becomes linear and the beam size can be approximated as

$$w(z) \approx w_0 \frac{z}{z_R}. \quad (1.14)$$

As a Gaussian beam travels propagates along the optical axis, it acquires an additional phase shift compared to a plane wave. This is known as the Gouy phase shift, $\phi(z)$, and it given by

$$\phi(z) = \tan^{-1} \frac{z}{z_R}. \quad (1.15)$$

At either side of it waist, the fundamental Gaussian mode will be shifted $\pm\pi/2$ relative to a plane wave. An alternative way of putting this is the fundamental Gaussian mode will gain a π change as it goes through from $-\infty$ to ∞ compared to a plane wave. Most of this phase change is gained within $\pm z_R$ of its waist. The

Gouy phase results in the the phase velocity at the waist being greater for a Gaussian beam than it is for plane waves.

The radius of curvature, $R_c(z)$, in terms of the Rayleigh range is

$$R = z + \frac{z_R^2}{z}. \quad (1.16)$$

For $z \ll z_R$ the radius of curvature can be approximated as infinite. As the beam approaches its Rayleigh length the radius of curvature decreases until it reaches its minimum at $z = z_R$. After this the radius of curvature begin to flatten out again, and for $z \gg z_R$ the radius of curvature can be approximated as z . The defocus, $S(z)$, of a beam is a more practical way of describing a beam as it is easier to compute the effects of a beam interacting with a focussing element of an optical layout in terms of the defocus than it is with the radius of curvature. The defocus is defined as the inverse of its radius of curvature

$$S(z) = \frac{1}{R(z)} \quad (1.17)$$

The fundamental Gaussian mode can be expressed in terms of a complex beam parameter, $q(z)$. This parameter is useful for calculating how a gaussian beam is transformed by an optical system. From this parameter, all the physical quantities related to a Gaussian beam can be computed. The complex beam parameter is defined as

$$q(z) = z + iz_R. \quad (1.18)$$

The inverse of equation 1.18 can be written in terms of the curvature and radius of the beam:

$$\frac{1}{q(z)} = \frac{1}{R(z)} - i \frac{\lambda}{\pi w^2(z)}. \quad (1.19)$$

Equation (1.11) can be re written in terms of this the complex beam parameter,

$$E(\mathbf{r}) = E_0 \sqrt{\frac{2}{\pi}} \frac{q_0}{w_0 q(z)} \exp(i\phi(z)) \exp\left(-ik \frac{x^2 + y^2}{2q(z)}\right) \exp(-ikz). \quad (1.20)$$

The radius of the beam, gouy phase, and the defocus at a point z along the optical axis can be calculated from $q(z)$ using

$$w(z) = \sqrt{\frac{\lambda}{-\pi \Im\left(\frac{1}{q(z)}\right)}}, \quad (1.21)$$

$$\phi(z) = \tan^{-1}\left(\frac{\Re(q(z))}{\Im(q(z))}\right), \quad (1.22)$$

$$S(z) = \Re\left(\frac{1}{q(z)}\right). \quad (1.23)$$

Mode matching integral

1.4.1 Astigmatism

If a beam has an elliptical rather than circular profile, then the beam is astigmatic. For simple astigmatism, a beam can be described with two complex Gaussian beam parameters, one for the yz plane and one for the xz plane. These are referred to as the tangential (xz) and the saggital (xy) planes. A gaussian beam with simple astigmatism will have a different waist size and location in the tangential and sagittal planes. This type of astigmatism arises due to gaussian beams passing through a

cylindrical lens or having a non-zero angle of incidence to a mirror.

An simple astigmatic gaussian beam can be split up into two orthogonal fields
(rewrite in terms of $u(\mathbf{r})$)

$$\begin{aligned}
E(\mathbf{r}) &= E(x, z)E(y, z) \\
&= E_0 \sqrt{\frac{2}{\pi}} \sqrt{\frac{q_{x0}}{w_{x0}q_x(z)}} \sqrt{\frac{q_{y0}}{w_{y0}q_y(z)}} \exp(i\phi_{Av}(z)) \\
&\times \exp\left(-ik\frac{x^2}{2q_x(z)} - ik\frac{y^2}{2q_y(z)}\right) \exp(-ikz). \tag{1.24}
\end{aligned}$$

where $\exp(i\phi_{Av}(z))$ is the average of the Gouy Phase in the saggital and tangential planes. The equations for waist size, equation (1.21), Gouy phase, equation (1.22) and defocus, equation (1.23), remain the same. A beam with simple astigmatism can be though of as two independent beams, and their interaction with the optical layout can be calculated independently.

1.5 Gaussian Beams Traversing optical Layouts

As a Gaussian beam propagates through an optical layout, the beam parameter that describes changes. The effect of simple optical components can be modelled with ABCD matrices, and an entire layout can be modelled by multiplying a series of ABCD matrices together. Components that introduce astigmatism to the beam can be modelled as having seperate ABCD matrices for the tangential and saggital planes. This formalism can also be used to compute cavity eigenmodes.

When a beam interacts with an optical component it is transformed from a mode described by q_1 to a new mode q_2 according to

$$q_2 = \frac{Aq_1 + B}{Cq_1 + D} \quad (1.25)$$

where $n_{1,2}$ are the refractive indices of the medium $q_{1,2}$ are in. The coefficients A, B, C, D are defined by a coefficient matrix $\mathbf{M}$,

$$\mathbf{M} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}. \quad (1.26)$$

To compute the coefficient matrix for a series of elements numbered 0 to n , the matrices for each individual element should be multiplied together

$$\mathbf{M} = \mathbf{M}_n \mathbf{M}_{n-1} \dots \mathbf{M}_0. \quad (1.27)$$

The ABCD matrices for various components is summarised in table 1.1.

1.6 Higher Order Modes and optical imperfections

The fundamental gaussian mode describes the electric field of a beam in an ideal optical system. Imperfections in the optical layout, such as misalignment and mode-mismatch, results in a different mode shape. If two modes are misaligned slightly, one will see a mode profile that looks like two spots instead of one. A small mode-mismatch results in a ring shaped mode profile. These imperfections in the layout and the resulting beam shapes can be modelled as the optical layout scattering light

Optical Component	ABCD Matrix
space	$\begin{pmatrix} 1 & d \\ 0 & 1 \end{pmatrix}$
Mirror or lens (normal incidence)	$\begin{pmatrix} 1 & 0 \\ -S & 1 \end{pmatrix}$
Mirror (AOI = α , Sagittal Plane)	$\begin{pmatrix} 1 & 0 \\ -\frac{S}{\cos \alpha} & 1 \end{pmatrix}$
Mirror (AOI = α , Tangential Plane)	$\begin{pmatrix} 1 & 0 \\ -S \cos \alpha & 1 \end{pmatrix}$

Table 1.1: Table of the ABCD matrices for simple optical components.

into higher-order optical modes.

There are many solutions to equation 1.10. One complete, orthonormal basis set of functions that can be used to construct solutions to 1.10 are the Hermite-Gaussian (HG) modes. These modes have xy symmetry, and appear as 'square' shaped. These are useful for modelling misalignments, as this type of imperfection can be naturally broken into xy coordinates. Mode-mismatches are best described by the $r\theta$ coordinates, and so its natural to express these in terms of another orthonormal basis set known as the Laguerre-Gauss (LG) modes. However, if the beam is astigmatic then the mode no longer has the symmetry required in order to be modelled by LG modes.

Equation (1.10) can be solved using HG modes, which have the mathematical form of

$$u_{nm}(x, y, z) = u_n(x, z)u_m(y, z). \quad (1.28)$$

The functions $u_{n,m}$ have the form

$$u_n(x, z) = \left(\frac{2}{\pi}\right)^{1/4} \left(\frac{\exp(i(2n+1)\phi(z))}{2^n n! w(z)}\right)^{1/2} H_n\left(\frac{\sqrt{2}x}{w(z)}\right) \exp\left(-i\frac{kx^2}{2R_c(z)} - i\frac{x^2}{w^2(z)}\right). \quad (1.29)$$

where $H_n(x)$ a Hermite polynomial of the order n . By writing $u_{nm}(x, y, z)$ out explicitly,

$$u_{nm}(x, y, z) = (2^{n+m-1} n! m! \pi)^{1/2} \frac{1}{w(z)} \exp(i(n+m+1)\phi(z)) \\ H_n\left(\frac{\sqrt{2}x}{w(z)}\right) H_m\left(\frac{\sqrt{2}y}{w(z)}\right) \exp\left(-ik\frac{x^2+y^2}{2R_c(z)} - i\frac{x^2+y^2}{w^2(z)}\right), \quad (1.30)$$

we can see that the higher order mode solutions to equation 1.10 have the same waist size and radius of curvature to the fundamental mode, but different intensity distributions and additional Gouy phase. If n and m are set to zero, then equation 1.11 is recovered. These HG modes interact with optical elements in the same way as the fundamental mode.

The eigen mode of a cavity is determined by its length and the radius of curvature of its end mirrors. The Gaussian mode that is the eigen mode of the cavity will have the same radius of curvature at the end mirrors of the cavity as the mirrors themselves. As higher order modes share the same radius of curvature as the fundamental

mode, these modes can also be resonant within the cavity. What separates these modes out in a cavity is the round trip Gouy phase that is acquired. Most cavities are designed such that the round trip Gouy phase is such that only one mode can be resonant at once, and all the other modes destructively interfere.

The Gouy phase separation of higher order modes is also important when designing alignment and mode matching control systems, as different types of errors have different phases. For example, angular and translational misalignment are both first order effects. This means they both transfer light from the HG_{00} mode into the HG_{10} . However, the complex amplitude of these misalignment situations is separated by $\pi/2$ rad of phase. Having two sensors separated by $\pi/2$ of Gouy phase, one can derive alignment control signals.

In general, a mode can be written as a superposition of HG modes

$$E(x, y, z) = \sum_{n=0}^{n=\infty} \sum_{m=0}^{m=\infty} A_m u(x) A_n u(y), \quad (1.31)$$

where A is the mode amplitude. This form is useful when calculating the coupling between modes.

The following example of the amount of power lost due to two arms of an interferometer being misaligned. The field at the AS port of the interferometer can be written as

$$E_{AS} = \frac{A_0}{\sqrt{2}} \left(\frac{\Delta A_0}{A_{00}} u_0 + \sum_{m=1}^{\infty} \frac{a_m u_m}{A_{00}} \right), \quad (1.32)$$

where A_0 is the incident field amplitude and A_{00} is the amplitude of the field in the arms. The change in the fundamental power due to an optical imperfection in one

of the arms is ΔA_0 . E_{AS} is normalised to the incoming power at the beam splitter. As the modes are orthogonal by calculating the power in the higher order modes, the power lost from the fundamental mode can be calculated. The amplitude of a mode can be written as

$$A_m = \frac{1}{\Gamma(m/2 + 1)2^m\sqrt{\pi}} \int_{-\infty}^{\infty} E(X)u_m(X)e^{-X^2/2}dX \quad (1.33)$$

where Γ is the Gamma function, X is the normalised distance $\sqrt{2}x/w$ and $E(X)$ is an arbitrary field distribution. For a misaligned beam with a purely angular misalignment θ , a linear phase gradient $\varphi(x) = 2\pi\theta x/\lambda$ must be applied to the regular field distribution. The field distribution at the waist can be written as

$$E(X) = \exp\left(iX\frac{\theta}{\theta_c} - \frac{X^2}{2}\right) \quad (1.34)$$

where the critical angle is $\theta_c = \lambda\sqrt{2}/\pi w$. Inserting this $E(X)$ into equation (1.33) gives

$$A_m = \frac{e^{-(\theta/\theta_c)^2}}{\Gamma(m/2 + 1)2^m\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-(-X+i\frac{\theta}{\theta_c})^2} u_m(X)dX, \quad (1.35)$$

which can be solved analytically by using the relationship

$$\int_{-\infty}^{\infty} e^{-(z^\sim t)^2} u_n(t)dt = 2^n \sqrt{\pi} z^n \quad (1.36)$$

to give

$$A_m = \frac{(i\frac{\theta}{\theta_c})^m e^{-(\theta/\theta_c)^2}}{\Gamma(m/2 + 1)}. \quad (1.37)$$

The fundamental order is reduced by

$$\frac{A_0}{A_{00}} = \exp \left((-\theta/\theta_c)^2 \right) \quad (1.38)$$

1.7 Concluding remarks

Chapter 2

Balanced Homodyne for LIGO A+

There doesn't need to be a discussion about what BHD is, but it is worth briefly mentioning what it is and the other readout methods again Context for BHD upgrade being part of A+.

2.1 BHD versus DC readout for A+

- Reduction of SRC length coupling. Should this be a quantitative analysis or shall I quote the results?
- Reduction of Back scatter from the septum window between ham5 and ham6. Technically this is not to do with BHD, this could, but practically it kind of is?
- Ability to tune homodyne phase is cool but in the context of GW detection, maybe not important.
- PD SNR is not important.

- There is an optomechanical coupling between SRC cavity length noise and GW signal channel, but as far as I can tell this is well below the noise floor.

some useful references:

- G2000962, DDR CDR presentation A.
- G2001026, DDR CDR presentation B.
- Ken's notes on BHD benefits compared to DC readout, email on 25/06/2020 11:26
- LIGO-L2000200 Q and A. Useful technical points and things people will find confusing/may ask about.

Main benefit of BHD is reduction of SRC length noise coupling to AS port. This noise coupling is described in [1], see equation 13. This paper has some coupled cavity maths in it. If this needs explained, look up Transfer Functions for Fields in a 3-Mirror Nested Cavity, [2]. There is an opto-mechanical coupling between the length of the SRC and arm cavity (due to radiation pressure). Reducing the DC offset reduces the radiation pressure, which reduces this coupling.

Estimate reduction of SRC length noise coupling Differential DC offset is ϵ , so this coupling scales with the offset. Since we get the LO from elsewhere, the amount of offset can be reduced. Currently it is TEM00 is decreased to fringe offset (rms + DC) + 1mW of common mode carrier, currently there is 20mA of photo-current. (shelia dwyer answer in email FW: BHD review committee request & document PREVIEW). Assume 0.85 A/W ($q_e = 99\%$), this gives 23mW of light.

This can be reduced to between 1mW to 5mW, giving a reduction of at most 23, at least 4.6. Need to check this point.

Optical spring effect which is porbably not important In a large scale gravitational wave detector, the main advantage of BHD versus DC-readout is the ability to reduce the differential arm offset that is required to provide the GW signal a local oscillator field to beat with, as the LO field is provided by other means. SRC length noise scales quadratically with the DC offset [1], so a reduction in the DC offset causes a reduction in this coupling. SRC length noise couples to the output signal via radiation pressure. The change in differential arm length due to a change in length of the SRC scales linearly with the DC offset ($\epsilon = kL_{\text{offset}}$) due to radiation pressure. Don't know what the current DC offset is. The optomechanical coupling between SRC length and differential arm length is given as

$$\frac{\partial L_-}{\partial l_s} = \frac{16g_s^2 r_a' r_s}{ct_s^2 s_\mu^2 (1 + s_{\text{rse}})} \epsilon k P_a \quad (2.1)$$

and the normalised response between the power at the AS port and the SRC length noise is

$$\frac{S_{\text{as}}}{S_{\text{dc}}} = 4g_p^2 g_s^2 r_a'^2 \epsilon \left(\frac{\partial L_-}{\partial l_s} \right) \frac{k \Delta l_s}{1 + s_{\text{rse}}} \quad (2.2)$$

2.2 Local Oscillator Path Length Stability Requirement and HRTS requirement

The requirement for the path length stability of the local oscillator is determined by the sensitivity goal for A+. Deconvolving the strain sensitivity curve for A+ with the linear response function between differential strain, h , and power at the AS port gives the noise of the AS beam. This is then converted to an equivalent phase noise between the AS the LO beams. Finally, the requirement for the suspended optics is derived.

A gravitational wave produces a phase modulated signal in the AS beam. The LO beam will also be phase modulated by the motion of the mirrors in the LO path. When these beams are combined, it is the phase difference between the two fields that produces a signal, so the phase noise of the LO path must be sufficiently low that the signal that is generated by the interference of the LO and AS beams corresponds to the phase modulation of the AS beam.

2.2.1 Local Oscillator Path

Two options for the local oscillator path were considered: a beam picked off from the main beam splitters AR coating (BSAR) or the beam transmitted through PR2 [3] (POP). The determining factor in choice of LO path was the effect of backscatter, and the POP choice is far superior to the BSAR choice at all frequencies and scatter phases. This is because any backscatter from the BSAR enters only one arm, and therefore is a differential effect. Any back scatter that enters the POP port is common-mode, meaning that the differential arm signal is far less sensitive to this

source of backscatter.

The layout for the optics which combine the the LO and AS beam is shown in figure 2.1. The beam transmitted by PR2 is used for control of the interferometer, so BHDBS1 is needed to pick off some of this light. The LO beam is steered towards the combining beam splitter BHDBS2. with BHDM1. Two lenses (**need names for these**) are used for mode matching the LO beam to the AS beam. The AS beam is transmitted through the SRM and OFI, and then is steered towards BHDBS2 with OM0.

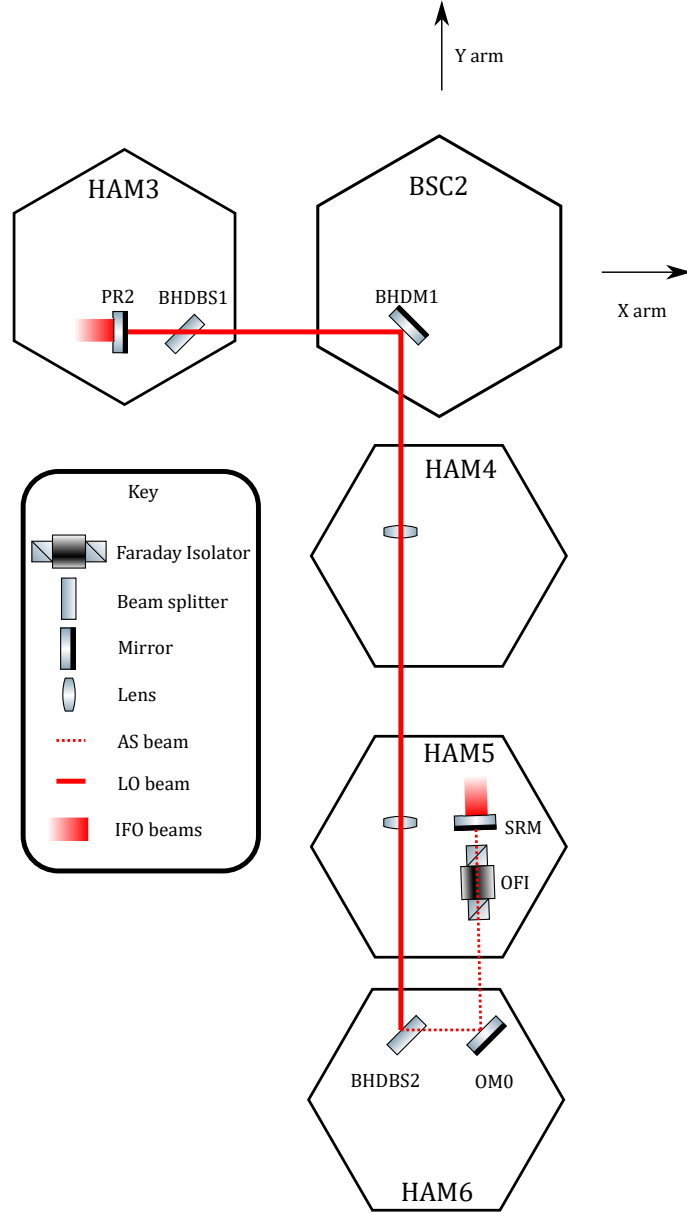


Figure 2.1: Sketch of the LO and AS beam paths. The LO beam is obtained from the light transmitted by PR2. Some of this light is needed for control of the interferometer, so BHDBS1 is used to split the light. The LO beam is then directed towards the combining beam splitter BHDBS2 with BHD1. Two lenses are needed to mode match the LO beam to the AS beam. The AS beam is transmitted through the SRM, passes through the OFI, and is directed to BHDBS2 via OM0.

2.2.2 Simulation of the Linear Response Function Between a Differential Strain and the Power at the AS Port

The linear response function of the A+ detectors was obtained using FINESSE. Since the core optics of the interferometer will not be changing in a way relevant to this calculation, the LLO model[4] was used to calculate the response between differential motion of the ETMs and the power transmitted through the OMC. To apply a differential arm signal with FINESSE, `fsig` commands were used to apply two signals in antiphase with each other to the ETMs,

```
fsig sig1 ETMXHR 1 0
fsig sig2 ETMYHR 1 180.
```

The frequency of this modulation was swept, and the signal produced at the AS port was measured by demodulating the light at the modulation frequency. The commands

```
xaxis sig1 f log 5 5000 2999
put P_DC_OMC_probe f1 $x1
```

do this. Since the input modulation is doubled by moving the mirrors differentially the response needs dividing by 2. To account for the ETMs retro-reflecting the light (or is the weird Finesse convention), the response needs dividing by 2 again. To convert this from phase units to strain units, a factor of $\frac{\lambda}{2\pi L_{\text{arm}}}$ is needed. Therefore, the output of the the FINESSE simulation needs to be multiplied by $\frac{1}{2} \frac{1}{2} \frac{2\pi L_{\text{arm}}}{\lambda}$ to be in units of $\frac{\text{W}}{\text{h}}$. The linear response function between a differential arm strain and the power at the AS port is shown in figure 2.2.

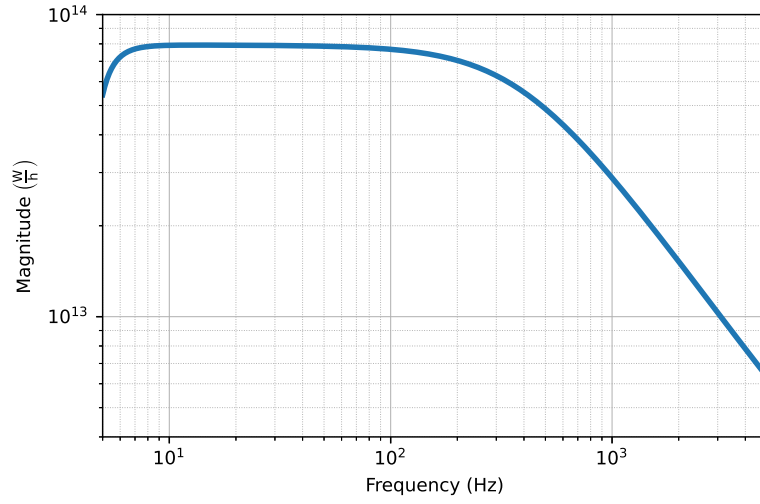


Figure 2.2: The linear response function between strain and light power at the AS port was obtained from a FINESSE simulation of the LLO detector[4]. The corner frequency at ≈ 300 Hz comes from the pole of the coupled SRC-arm cavity. The behaviour below 6 Hz is due to optical spring effects.

2.2.3 Converting the Sensitivity Curve for A+ into the HRTS requirement

The sensitivity curve will be shown in an earlier chapter. To convert the A+ sensitivity curve into a path length stability requirement for the local oscillator, a few steps are required. First, the sensitivity curve must be expressed in units of $\frac{W}{\sqrt{\text{Hz}}}$. Secondly, the intensity noise of the AS beam needs to be converted into a phase noise. In order for the phase noise of the local oscillator to not limit the performance of the gravitational wave detector, the phase noise of the AS beam must be greater than the phase noise of the LO beam. Finally, a factor of 10 safety margin is required. The HRTS requirement is then obtained by including the number of reflective optics in the LO path.

The A+ sensitivity [5] curve was multiplied by the function shown in 2.2 to convert it from units of $\frac{h}{\sqrt{\text{Hz}}}$ to $\frac{W}{\sqrt{\text{Hz}}}$. This is shown in figure 2.3. At frequencies above 1 kHz, the requirement is flat as this is the region dominated by shot noise. At lower frequencies, the requirement is relaxed as the sensitivity curve becomes dominated by other noise sources.

The quantum mechanics of this problem have been included in the A+ sensitivity curve [5], therefore we can approach this with classical physics. When the LO and AS beams interfere at the beam splitter (BHSBS2), the power of the resulting beam is

$$P = |E_s e^{i\omega t} + E_{\text{LO}} e^{i\omega t + \phi}|^2 = |E_s|^2 + |E_{\text{LO}}|^2 + 2|E_s E_{\text{LO}}^\dagger| \cos(\phi), \quad (2.3)$$

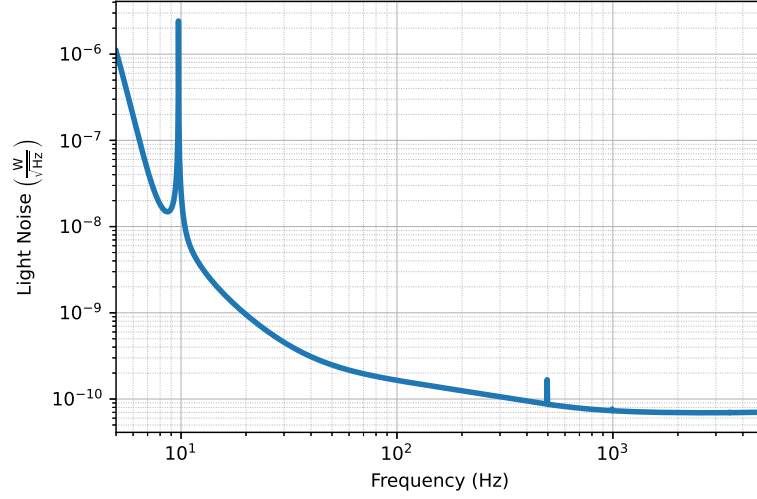


Figure 2.3: The intensity noise of the light at the AS port for the A+ detector. This was obtained by deconvolving the detectors sensitivity curve[5] with the function shown in figure 2.2.

where E_s and E_{LO} are the (complex) field amplitude of the signal and local oscillator beams, ω is the laser frequency, t is time, and ϕ is the phase difference between the local oscillator and the signal beam, also known as the homodyne angle. Differentiating with respect to ϕ gives the relationship between a change in power and a change in phase between the two beams,

$$dP = -2|E_s E_{LO}^\dagger| \sin(\phi) d\phi. \quad (2.4)$$

and since the homodyne angle is 90° , this becomes

$$dP = -2|E_s E_{LO}^\dagger| d\phi. \quad (2.5)$$

To convert ϕ to distance, use

$$\frac{\lambda}{2\pi}d\phi = dx. \quad (2.6)$$

The light noise curve shown in figure 2.3 should be multiplied by $\frac{\lambda}{2\pi} \frac{1}{2|E_s E_{\text{sig}}^\dagger|}$ to obtain the the noise floor of the interferometer in units of LO path length noise. For a 98 mW LO beam and a 5 mW AS beam, the noise floor of the interferometer in units of LO path length is shown in figure 2.5.

The HRTS requirement assumes, based on the current design, that there will be 2 mirrors at 45° which are used to steer the AS and LO beams. As the optics reflected at 45 degrees, the path length change due to the optics motion is multiplied by $\frac{2}{\sqrt{2}} = \sqrt{2}$. The beams are combined on a beamsplitter, and the notion of this beam splitter needs to be counted twice. A sketch of this is shown in figure 2.4. A factor of 10 is then included as a safety margin since there are many uncorrelated noise sources in LIGO. With a 98 mW LO beam and a 5 mW AS beam, the requirement for the HRTS optics longitudinal motion at 100 Hz is $5 \times 10^{-17} \text{m}/\sqrt{\text{Hz}}$. A typical motion of the optical benches at 10 Hz is $10 \times 10^{-1} \text{m}/\sqrt{\text{Hz}}$ [6]. To achieve this with passive isolation, a triple suspensions will be used. make figure showing requirement as a fuction of AS power?

2.2.4 HRTS

remove references to HRTS in previous section. Refer to the HRTS there as table motion isolation requirements, or something like that. Discuss the HRTS in its own separate section.

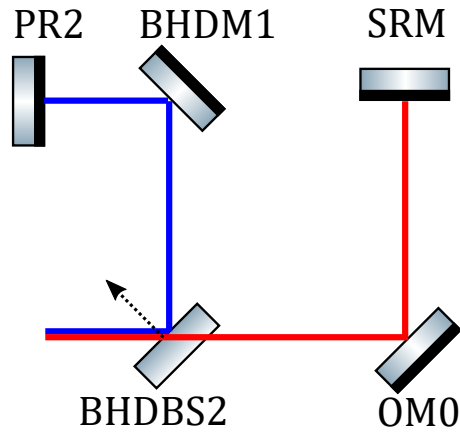


Figure 2.4: A sketch of the LO and AS beam paths. The LO beam path is shown in blue and the AS beam is shown in red. A dashed line indicates the direction that the recombining beamsplitter, BHDBS2, moves. When this happens, at the point of the beams combining the phase of the AS beam has increased while the phase of the LO beam has decreased. This means that the recombination beamsplitter motion needs to be counted twice.

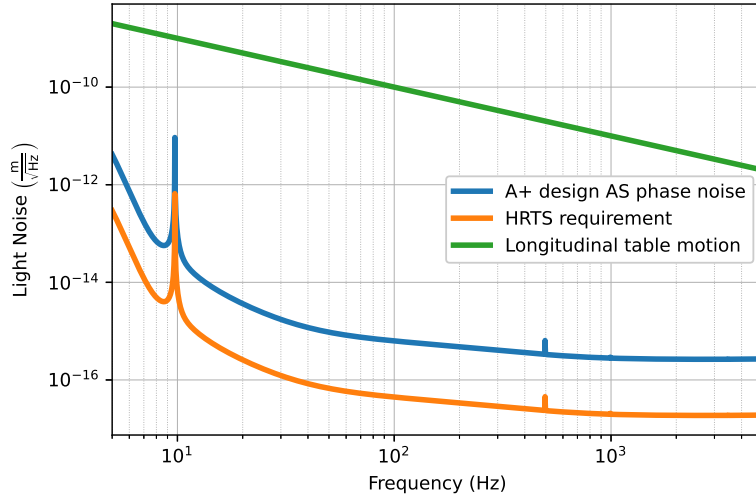


Figure 2.5: The phase noise between the LO and the AS beams causes noise. The sensitivity curve for A+ sets the limit on how much path length noise is acceptable between the AS and LO beams. The blue line shows sensitivity curve is shown in units of path length noise between the AS and LO beams, and the orange line shows the requirement of HRTS mirror motion.

2.3 Alignment of the LO and AS beam

2.4 HAM6 layout

Fundamentally, the layout in HAM6 must combine the AS beam and the LO beam, and sense the combined beams to form a balanced homodyne detector. There are several other functionalities required and constraints imposed on the layout of HAM6. The following section will describe these functionalities and constraints and how they guided the design of the HAM6 layout. This work was undertaken by a team including myself, Ken Strain, Russel Jones, Stephen Webster and Mark Barton.

2.4.1 The Constraints and Requirements of the Layout of HAM6

The entry point of the AS and LO beams in principle can be anywhere in HAM6, however in reality they are fixed due to the location of HAM6 with respect to the interferometer, due to the location of the septum windows between HAM5 and HAM6, and due to the many other components of the interferometer that these beams need to avoid. The LO beam is steered with BHDM1 onto BHDBS2, and the AS beam is steered with OM0 onto BHDBS2. This is required to provide the degrees of freedom to co-align the AS and LO beams. There is some freedom in choice for the location of OM0 and BHDBS2, but there is no choice in the location of the AS and LO beams.

The design beam parameter at the SRM is $q_{\text{SRM}} = -3.55 + 1.09i$ m, so the design of the HAM6 mode matching optics is based on this number. In reality, the beam parameter could be different to this, as discussed in chapter 3). At the SRM, there are two sets of modes we could design the AWC to match to: The arm mode, or the SRC mode. Ideally, these would be the same mode, however geometry errors in the SRC cause the arm and the SRC to be mismatched meaning the radius of curvature and beam size at the SRM will not correspond to the eigenmode of the SRM. This set of likely modes, shown in 3.11, was taken into consideration when choosing a layout for the AWC in HAM6.

After BHDBS2, the layout the optics between OMC input couplers and BHDBS2 must provide the functionality for steering the combined AS and LO beam into the OMC, and provide mode matching for the AS beam to the OMC. In addition to this, sensors in HAM6 are required for targeting the AS and LO beams. Some of these functions will be achieved using beams which are picked off from the main beam path, and so additional mirrors will be required to steer these beams around the chamber. A set of smaller optics will be installed to provide diagnostic beams which can be imaged with CCDs. A shutter will be required to protect the hardware in HAM6 from being damaged by the large amounts of light that can potentially be transmitted through the SRM.

After the AS and LO beams are combined, double stage suspensions will guide the beam from BHDBS2 to the OMC. These suspensions provide active positioning control and active wave front control. In principle, the role of the AWC in HAM6 is to correct for geometry errors of the optics after the SRM such that the mode

matching between the Arm mode and the OMC waist is maximised. This assumes that the changes in the interferometers mode, for instance due to a change in thermal state, will be corrected for by other subsystems e.g. the TCS. In practice, operators of the interferometer do what ever improves the SNR of the the gravitational wave signal. For instance, it has been found that TCS can be used to decrease laser intensity noise coupling to the the GW signal channel **CHECK**. The AWC in HAM6 was designed with this flexibility in mind. The losses due to components in HAM6 is summarised in table 2.1 (See E1900385).

To provide orthogonal alignment control in each beam path after BHDBS2, two of the mirrors are required to be separated by 90° of Gouy phase. Additionally, to provide orthogonal mode matching control, two of the mirrors should be separated by 45° . The Gouy phase separation requirements give us a starting point for designing layouts, as these can be interpreted as needing the mode matching mirrors to be as far apart as possible from each other on the HAM table. As the Rayleigh range of the beams is on the order of a meter, and most gouy phase is accumulated within one Rayleigh range either side of a beam waist, a beam will have to travel quite far across the table before it acquires sufficient Gouy phase. While in principle only two curved mirrors would be required to mode match the beam from the interferometer to the OMC, in fact a single mirror may be able to do that, a lens in the OFI will be used to provide further Gouy phase separation between the two active mode matching mirrors after BHDBS2.

The goal for mode matching between the arm mode and the OMC mode is is $> 98\%$. The SAMS actuators will provide ± 100 , mD of optical power tuning range,

with a bandwidth of the order 100 Hz. Astigmatism can be considered as a loss, and non-zero angles of incidence (AOI) of beams onto curved optics results in the beam becoming astigmatic. This prevents designs using large AOI. This is described in detail in Chapter 1.

The current OMC design (with trivial modifications) will be used to remove junk light from the signal beam, so there is no choice in target waist size. The input beam at the SRM is defined in T1200410. The mode matching is designed assuming that the beam at the SRM is the ‘design mode’. Mode matching between the beam exiting the SRM and the OMC waist will be done with a static lens and 2 active mirrors. The static lens will be placed just after the OFI and its purpose is to provide sufficient Gouy phase separation of the 2 active mirrors.

The light transmitted by the OMCs is used to measure the gravitational wave signal, since the higher order modes are suppressed by a factor of the finesse of the OMC. The OMCs are expected to be the largest contributor of back-scatter from HAM6 back to the interferometer. By having the OMCs after the AS and LO beams are combined, and by having the OFI protect the interferometer, the requirement on the motion of the OMCs is fairly lax. Nevertheless, a single stage suspended platform is required. This will provide isolation down to 4 Hz.

2.4.2 Chosen Layout

Optically, there are many layouts which could have satisfied the above requirements and constraints. It was decided that the two beams after BHDBS2 would have the same path length to reduce the cost of the HAM6 optics. Ultimately the choice

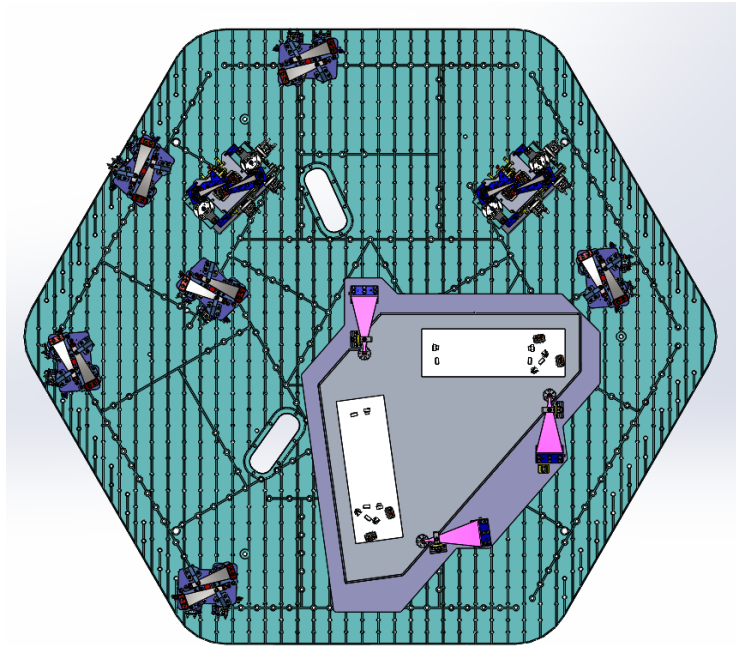


Figure 2.6: Place holder graphic. The pentagram layout. CAD will be inserted later.

HAM6 loss Source	Target for O5
OMC throughput	$< 2\%$
OM3 transmission (or equivalent) for sensing	1–2%
PD quantum efficiency	1%
Mode mismatch to the OMC	$< 2\%$ (as low as possible)
Any other optical loss	$< 1\%$

Table 2.1: Target for losses in HAM6. Reproduction of E1900385

Space Name	Optical Path Length (mm)
SRM to OFI lens	1120
OFI lens to HAM6 table edge	720
HAM6 table edge to OM0	432
OM0 to BHDBS2	905
BHDBS2 to OMxS	225
OMxS to OMx1	1240
OMx1 to OMx2	1240
OMx2 to OMCx waist	1240

Table 2.2: Spacings for the main beam path from the SRM to the OMC waist.

of layout for HAM6 was determined by mechanical factors such as table space and number of allowable suspensions. The design that is being taken forward at the time of writing is known as the ‘pentagram’, and is shown in figure 2.6.

Show plots produced by `ligo_layout.py`

2.4.3 WS visulations as a tool for understanding mode matching problems

Hermite Gaussian beams will be explained earlier in the intro chapters, so don’t need generic expansion, just use the 2nd order ones. Have a read of [10].

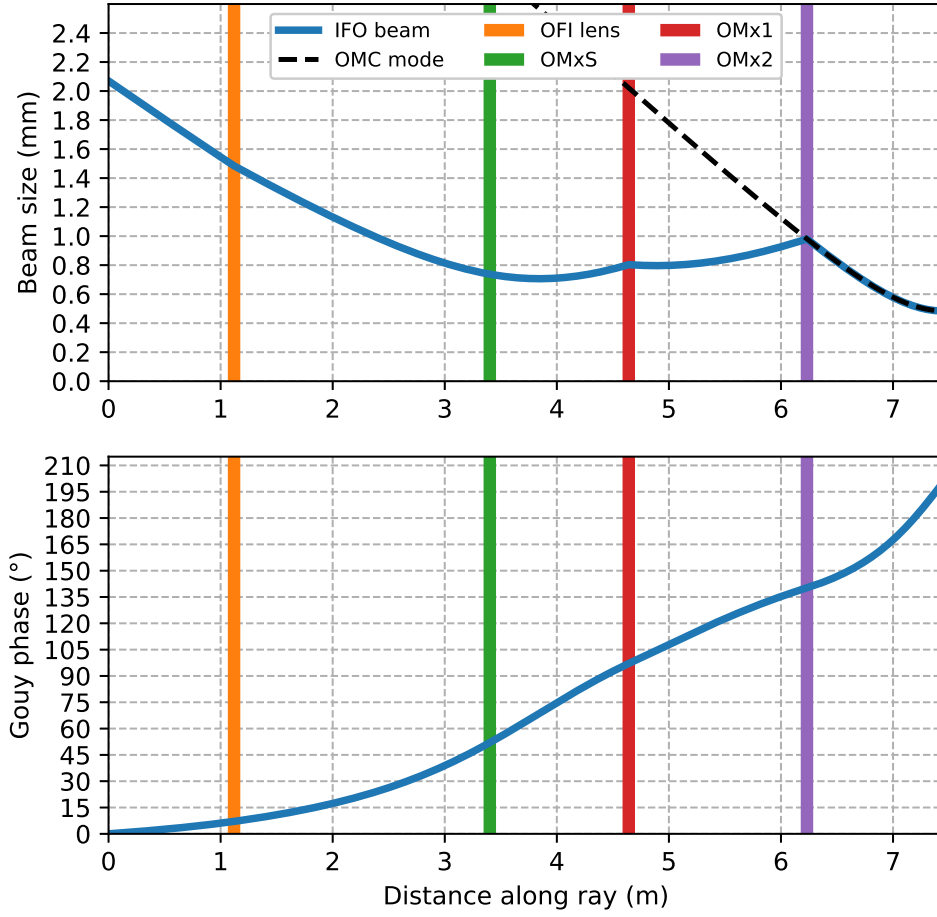


Figure 2.7: The radius of the beam as it travels from the SRM ($z = 0$) to the waist of the OMC.

Optic Name	Distance along optical axis (m)
SRM	0.000
OFI lens	1.122
OM0	2.272
BHDBS2	3.177
OMxS	3.402
OMx1	4.642
OMx2	6.232
OMCx Waist	7.452

Table 2.3: A table of the position along the optical axis for each optic between the SRM and the OMC waist. Since there are two identical beam paths after BHDBS2, path A and path B, the names of OMA1, OMB1 etc. have been written as OMx1 etc.

The mode-matching between two laser beam modes, or a laser mode and a cavity mode, describes how similar those two modes are. In general, the two modes can be mismatched by being misaligned or by having a different beam shape. Mode matching typically refers to the overlap of the two modes if they were perfectly aligned.

Mode-matching is usually, to first approximation, thought of as a difference in beam waist size and beam size location. This can be mathematically described by the $HG_{20} + HG_{02}$ mode. The effect of changing the waist location of the mode is equivalent to an in-phase change in HG_{20} content, whereas a change in waist location is orthogonal to this. [read up on this](#). Sensing the $HG_{20} + HG_{02}$ mode therefore gives a measure of the mode-matching between two lasers. As the $HG_{20} + HG_{02}$ acquires 3ϕ compared to a plane wave, and 2ϕ compared to the HG_{00} mode, spacing the actuators at 45° will give orthogonal control of the mode matching. This

means that the beam size and beam waist can be independently controlled.

A curved mirror can only change the curvature of a Gaussian beam, meaning that at the mirrors surface the beams before and after reflection must have the same size. The process of a Gaussian beam described by q_1 being reflected from a mirror with power S_m to produce a new Gaussian beam, q_2 , can be written as

$$\frac{1}{q_2} = S_m + \frac{1}{q_1} = (S_m + S_1) - i \frac{\lambda}{\pi w_1^2}. \quad (2.7)$$

To visualise how a mode is changed by an optical layout, beam parameters can be represented in a 'WS space'. This space parametrises a beam in terms of its size, w , and its defocus, S . This has the advantage of showing the beam in terms of practical quantities. The downside to this representation is as the beam propagates through the layout, the path through the WS space is fairly complex. There are two other ways that a beam parameter could be graphically represented. As q is a complex number, the beam parameter could be represented in terms of its real and imaginary components. This makes seeing the effect of propagation straight forward, but the effect of a focussing element is harder to visualise. Alternatively, plotting $1/q$ in terms of $\lambda/\pi w^2$ and s produces plots where the effect of a focussing element and propagation is easy to visualise, as focussing is just a change in defocus and propagation follows a circular path. However, this technique obscures the beam size, and as these graphs are being used to visualise the beam, rather than design the optical layout, the WS representation was chosen.

To demonstrate how orthogonal actuators can be represented in WS space, consider the two lenses with variable power separated by 45° of Gouy phase. As the two

lenses are orthogonal, the waist size and waist location of the mode exiting this layout will orthogonally be controllable. In a WS space, the orthogonality of two actuators can be seen by the angle between the two vectors that represent the actuation. If the two lenses are orthogonal, then the actuation vectors will be orthogonal. Figure 2.8 is a sketch of the WS space representation for a simple optical layout consisting of two active lenses separated by 45° Gouy phase. If there is large amounts of astigmatism, it no longer makes sense to represent the beam with just one WS space diagram.

Two perfectly mode matched beams will have identical beam parameters. If the input mode and the target mode differ in beam size and defocus, then some light will be scattered into the $\text{HG}_{20} + \text{rmHG}_{02}$ mode. The mode matching can be calculated at any point along the optical axis of the layout, as the mode matching must be preserved, but it makes sense to choose a physically significant plane such as the input plane or the plane containing the waist of the target beam. The amount of mode matching $\mathcal{M}$ between two astigmatic modes, $q_{1,2}$, is

$$\mathcal{M} = \frac{\sqrt{(q_{x1} - q_{x1}^*)(q_{y1} - q_{y1}^*)(q_{x2} - q_{x2}^*)(q_{y2} - q_{y2}^*)}}{|q_{x2} - q_{x1}^*||q_{y2} - q_{y1}^*|}. \quad (2.8)$$

It is useful to plot contours of the mode overlap in the WS space between a mode and a target mode. Note that these contours can only tell you about the overlap between a mode and the target mode, the mode matching between two arbitrary modes can not be deduced from these contours. An example of a situation where this is obvious is two different modes may be, say, 90% mode matched to the target. As they are different from each other, they cannot be 100% mode matched.

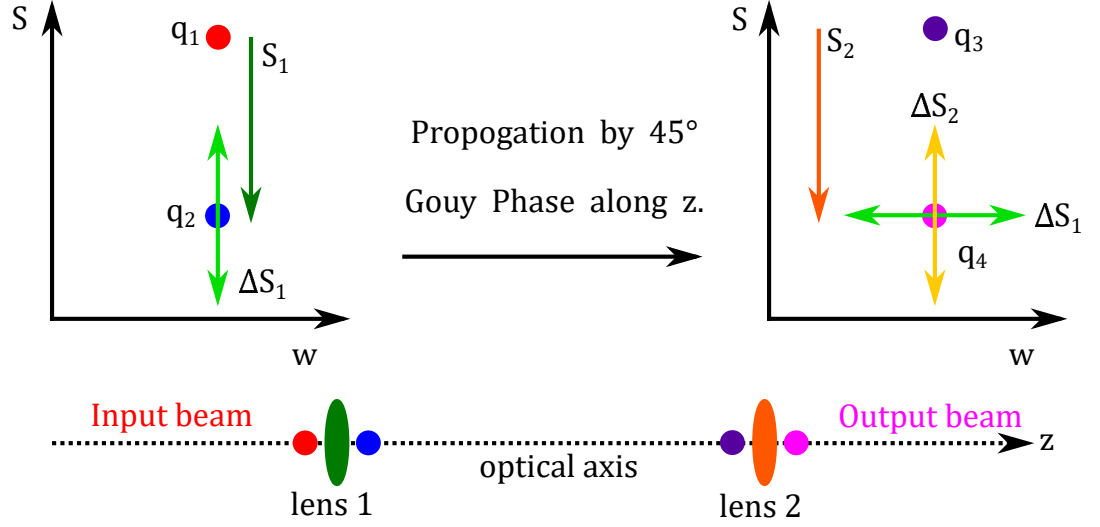


Figure 2.8: Sketches of the WS representation for a Gaussian beam interacting with different parts of a simple optical layout consisting of two active lenses with powers $S_{1,2} + \Delta S_{1,2}$. The left hand axis shows the WS plane at lens 1. The input beam q_1 is indicated by a red dot, and the effect of the lens on q_1 is shown by the dark green arrow S_1 . This results in a new mode q_2 , which is shown as a blue dot. A light green arrow indicates the actuation of lens 1. The right hand axis shows the WS plane at lens 2. The input mode at lens 2, q_3 , is shown as a purple dot, and the output mode, q_4 , is shown as a pink dot. The static and dynamic power of lens 2 is indicated by the orange and yellow lines. As lens 1 and lens 2 are separated by 45° of Gouy Phase, the actuation provided by lens 1 results in a change in the beam parameter which is orthogonal to the actuation provided by lens 2, provided that the actuations are small enough.

The mode matching requirement for LIGO is $> 98\%$. To visualise the effectiveness of the actuation provided by the active elements of the HAM6 layout, the region of the WS from which the target mode matching space achieved was computed. To do this, first the ring of modes in WS space that is 98% matched after reflection from the second active mirror, OMx2, was considered. These modes must be within $S_{\text{OMx2}} \pm \Delta S$, where S_{OMx2} is the static power of OMx2 and ΔS is the actuation range of the SAMS. If OMx2 had no actuation, then the effect of OMx2 would be to move the ring vertically in the WS plane. However, as power of OMx2 can be increased or decreased, the shape of the ring of modes that can be mode matched to 98% can be changed. Crudely, the shape of this space can be seen by applying $S_{\text{OMx2}} \pm \Delta S$ to the 98% ring, producing 2 new rings. If the ring is cut by a line which intersects the locations on the ring at the extrema of the beam size, then two regions of the 98% ring can be created. For one of these regions, to maximise the amount of WS space covered by the actuator, the actuation applied should be $+\Delta S$, whereas for the other region, the actuation should be $-\Delta S$. This gives a set of modes which can be propagated to the first mirror, OMx1, where the same procedure can be applied. This process shown in Figure 2.9.

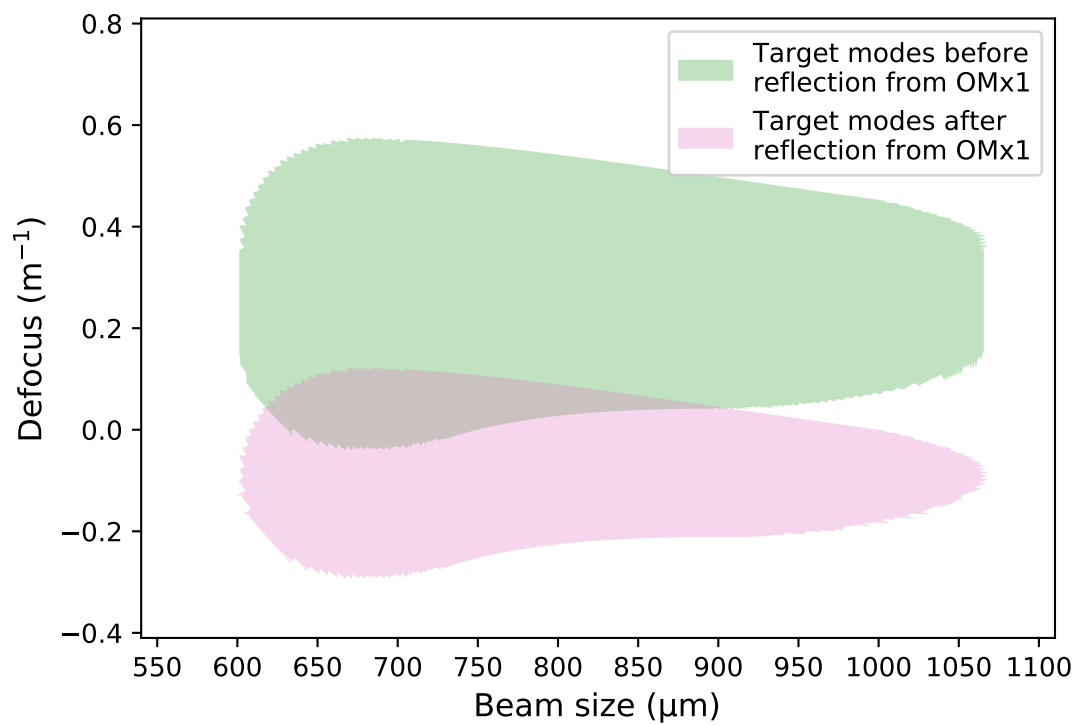
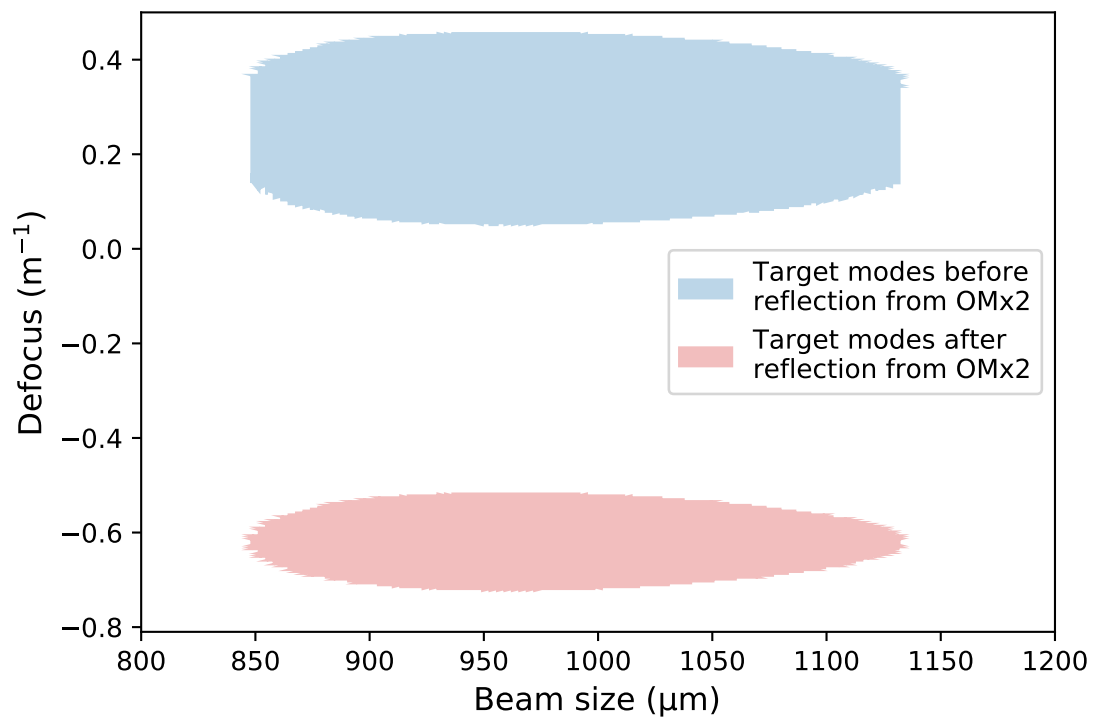


Figure 2.9 (*previous page*): Top panel: This is the WS space at the OMx2 plane. After reflection from OMx2, a mode must be at least 98% mode matched to the waist as there are no further active mode matching elements in the layout of HAM6. The target modes are shown by the red region. The blue region represents which modes can be reached from the red region by either $S_{\text{OMx2}} \pm \Delta S$, and was constructed by applying transformations to the modes in the red region. The red region was split into two parts, ‘upper’ and ‘lower’, which approximately corresponds to modes with a defocus greater than 0.6 or less than 0.6. The line that splits the red area into an upper and lower region corresponds to the line that connects the two extremes in width of the region, and does not have to be a horizontal line. To create the lower edge of the blue region, modes in the lower edge of the red region had the transformation corresponding to a mirror power of $S_{\text{OMx2}} - \Delta S$ applied to them. To construct the upper edge of the blue region, upper edge of the red region had the transformation corresponding to a mirror power of $S_{\text{OMx2}} + \Delta S$ applied to them. These regions were then joined by a vertical line that connects them.

Bottom panel: This is the WS space at the OMx1 plane. The pink mode represent the target modes that a mode must be able to reach by being reflected from OMx1. The blue region from the upper panel was propagated back to OMx1 to form the pink region. In a similar way as before, the pink region was split into an upper and lower region, and the green region was constructed from this. Modes in the green region are capable of being transformed to the pink region, which therefore means they can be transformed to the red region, meaning they satisfy the 98% mode matching requirement. The green region was propagated back to the through the layout to SRM, and this is as the purple region in Figure 2.10.

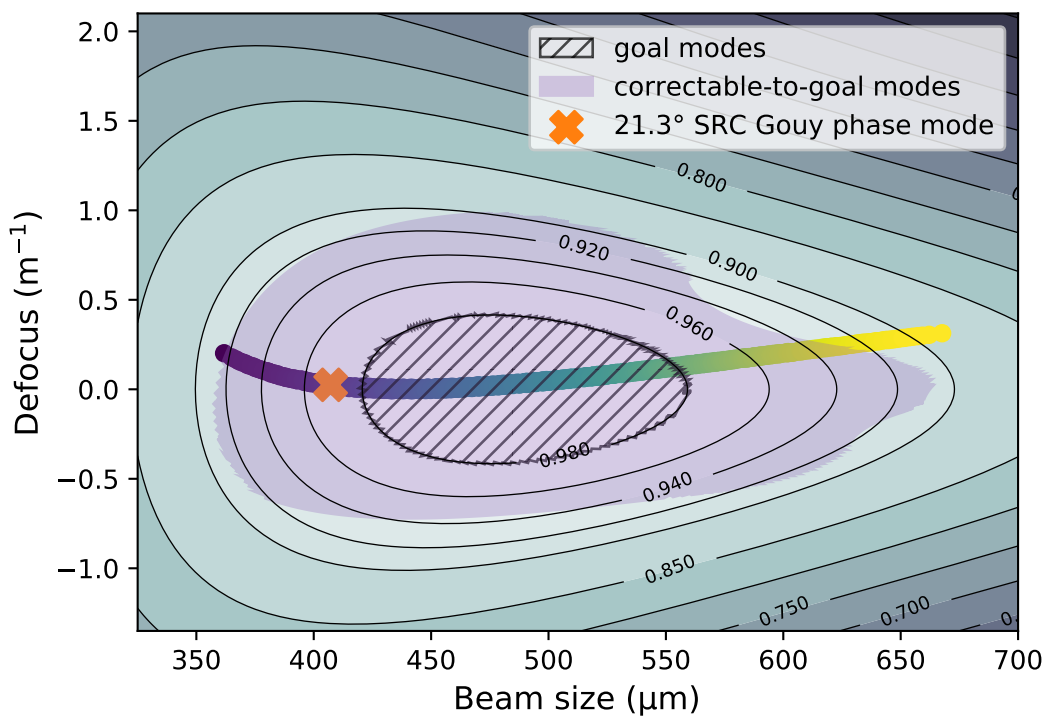
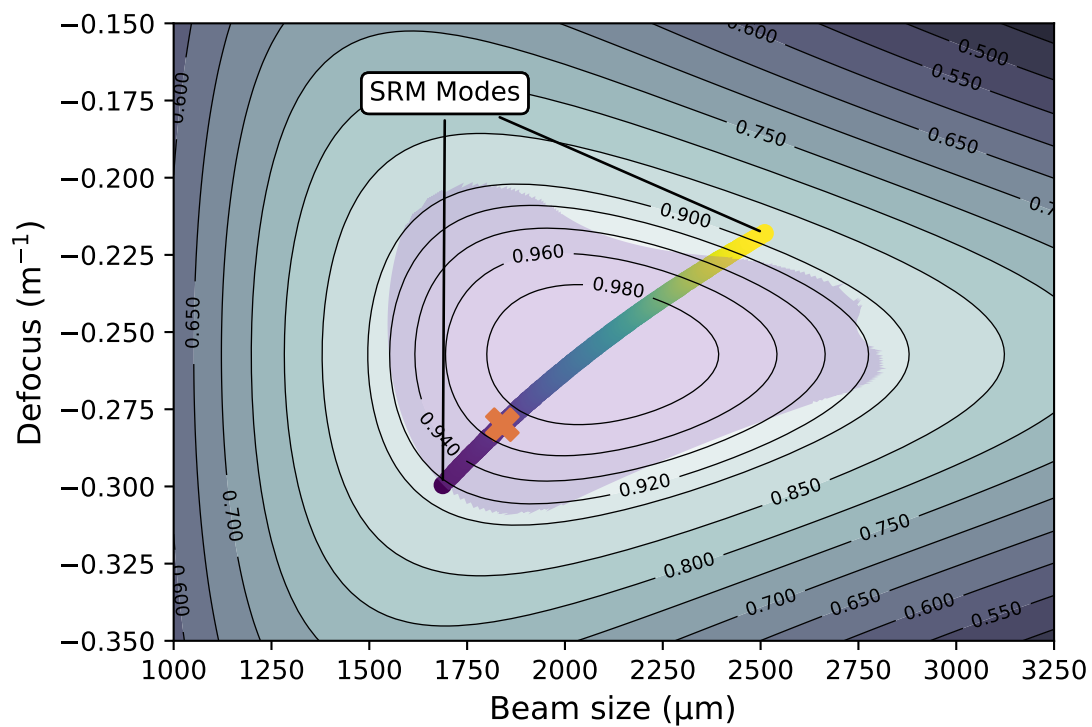


Figure 2.10 (*previous page*): Top panel: the set of modes that could be present at the SRM is shown by the purple-blue-green-yellow curve, as described in chapter 3. The purple region shows which modes can reach 98% mode matching by using the the AWC in HAM6. The orange cross indicates the mode that corresponds to the SRC having a single pass Gouy phase of 21.3° . The contours indicate the mode matching between the arm mode and the OMC mode. Bottom panel: this shows the same modes as the top panel, except they have been transformed to the plane of the OMC waist. The target region is highlighted with a hatched area.

The modes from the SRM that can be matched at to the target is shown in Figure 2.10. The visualise how successful the layout is at the handling likely modes from the arm described in Chapter 3, these modes were compared to the region of WS phase space from which the 98% mode matching requirement from a starting point of 92% mismatch. It was found that almost all likely modes will be covered by the layout, as the modes near the edge of this continuum are more unlikely to occur than the ones near the centre. For all layouts analysed, there were no major difference in the amount of modes that could be mode matched. Its worth noting that the in figure 2.10, modes at the yellow end of the spectrum may not be physical as they correspond to an unstable SRC. In principle, the requirement for having 45° of Gouy Phase separation between OMx1 and OMx2 doesn't maximise the amount of interferometer modes that can be mode matched. The shape that the purple region will have when HAM6 is constructed will depend on the optical powers of OMx1 and OMx2,

Nevertheless, Figure 2.10 is encouraging since it shows that the actuation provided by the OMx1 and OMx2 is very forgiving, so much so that the strict definition of the role of AWC in HAM6 may be relaxed in order for operators to use the AWC here to correct for other errors in the geometry of LIGO.

2.5 Conclusion about BHD for A+

Chapter 3

The Effect of Radius of Curvature Errors in SR3 and SR2 on the Beam Parameter of the Arm Cavity Mode at the SRM.

3.1 Introduction

A gravitational wave interacting with the LIGO detectors creates a differential signal in the arm cavities, and as the signal is differential it is directed to the Signal Recycling Cavity (SRC). The SRC tuning shapes the frequency response of a differential arm length change to light power transmitted through the antisymmetric (AS) port, with there being two modes of operation: signal recycling (SR) and resonant sideband extraction (RSE). The beam is then directed to the Output Mode Cleaner (OMC) to filter away ‘junk’ light. Junk light is comprised of higher order

modes (both spatial and temporal), and some of this junk light comes from mode mismatches between the arm cavities and the SRC. The light transmitted by the OMC ideally will only contain the signal generated by the differential arm length change, which is a signal that originated in the arm cavities. **This will be explained in the ‘detector’ chapter, so does this need references?**

Losses between the arm cavities and the readout photo-diodes result in less signal as well as more noise. The decrease in signal is straight forward to understand: if there are less photons to measure, then the signal will be smaller. Since the power is lower, the signal to noise ratio will be larger since shot noise goes as $\sqrt{\text{Power}}$. As gravitational wave detectors implement squeezing to decrease the quantum noise in the detection quadrature, it is important to preserve the quantum state of the signal field. This means that losses contribute additional noise, as losses pollute the squeezed quantum state with unsqueezed vacuum. **This type of loss will be featured earlier in the thesis in the ‘detector’ chapter, so does this need references?**

It is known that 20% of the light is lost between the AS port and the readout photo-diodes, as the shot noise can be compared to the expected shot noise for a given fringe slope, input power and known losses. Roughly half of the loss is accounted for e.g. some comes from the Faraday isolator, some comes from the OM3 pick off, however the other half is not fully understood e.g. see [12]. The level of squeezing can also be used to cross check the loss [13]. Some of the unaccounted for loss may come from mode mismatch between the arm cavities and the OMC.

Mode matching losses between the squeezed light source, the interferometer and the OMC act coherently. Mode mismatch can be modelled as a modal beam split-

ter that transfers some light from the HG00 mode to the HG20+02 mode. The amount of light transferred between the HG00 and HG20+02 mode is determined by the magnitude of the mismatch, and the phase difference between the HG00 and HG20+02 mode depends on the nature of the mismatch. An error in waist size is in-phase (real), whereas an error in waist location is out-of-phase (imaginary). In general the mismatch introduces a complex phase. A sequence of mode mismatches can be modelled in the same way as a Mach-Zehnder interferometer. The difference in phase between the HG00 and HG20+02 mode is analogous to a difference in path length in a Mach-Zehnder interferometer. It differs from a typical Mach-Zehnder interferometer as the HG00 to HG20+02 splitting ratio is not well balanced. As the mode matching losses act coherently, the degradation of the squeezed state can be greater for mode matching losses than for other types of loss e.g. the loss due to photo-diodes with a quantum efficiency less than one. If the squeezer is perfectly matched to the OMC, then the mode mismatch between the squeezer and the interferometer is four times the loss you would expect from treating a mismatch loss as a regular loss. This is explained in detail in [14].

Since the SNR of the gravitational wave signal is directly related to mode-matching between the arm cavities and the OMC, knowledge of the beam parameter at the SRM is highly important. The mode in the arm is well defined since the arm cavities are high finesse. Uncertainties in the RoC of optics in the SRC telescope (SR2 and SR3) results in an uncertainty of the beam parameter of the arm mode at the SRM, as well as the eigenmode of the SRC. A fair amount of mode mismatch can be tolerated by the SRC as it is a low finesse cavity, so this leads to an uncertainty in the beam parameter that is transmitted through the SRM.

Junk light will be generated if the arms are perfectly matched to the OMC but not to the SRC. If the arms are matched to the OMC but not to the SRC, then actuating on the beam parameter with the ITM thermal lens and the SRM thermal lens would not increase the signal as these two actuators are separated by enough Gouy phase that a large portion of the beam parameter phase space can be covered with them. This means the mode matching between the arm and the OMC can be optimal but the light can still contain HOM. Additionally, if the beam is astigmatic, then optimal mode matching would not correspond to perfect mode matching.

The aim of this work is to see how the errors in the focal length of SR3 and SR2 effect the beam parameter of the arm mode at the SRM and higher order mode (HOM) content of the beam transmitted through the SRM. A LIGO-like interferometer was simulated using FINESSE. The simulated interferometer has properties very close to the the as built LLO LIGO detector, however some simplifications were made to make the effect that a geometry error in the SRC has on the beam transmitted through the SRM clearer. The difference in the HOM content of the beam transmitted through the SRM between an ITM single bounce measurement and a measurement which involves the arm cavities was investigated. The effect that the interferometer being in signal recycling and resonant sideband extraction (RSE) mode had on the HOM content of the beam as the focal length of SR3 deviated from its nominal value was also simulated.

It was found that the dependence of the ratio of HG00 light to the total light depends mainly on the focal length of SR3 and is not effected by the presence of arm cavities. This is because the arm cavities are well matched to the input laser[15].

When the arm cavities are on resonance, the ITMs act as highly reflective mirrors with frequency dependent behaviour. This means that there is no difference in HOM content of the beam transmitted through the SRM between a single bounce measurement and an arm cavity measurement.

The HOM content of the light does depend on the detuning of the SRM. It does not depend on whether the interferometer is in signal recycling mode or RSE mode (as the arm cavities do not matter). When the SRC is on resonance, it is much more sensitive to an error in the SR3's focal length than when the SRC is anti-resonant. The light is less sensitive to an error in the focal length of SR2.

The beam parameter of the arm mode at the SRM behaves in a predictable way since the arm mode is well defined, so the Gouy phase of a beam at the SRM can be used to acquire information about the arm mode at the SRM. The beam width and RoC were simulated for a range of SR3 focal length errors. By analysing where the SRC becomes unstable, the beam size cannot exceed 2.2 mm. These results can be used to constrain design tasks related to active wavefront control. The Gouy phase has been measured to be $21.3 \pm 0.3^\circ$ [16]. If the thermal lensing in the ITM is roughly at design, then the focal length of SR2 is -3.205 ± 0.003 m and the focal length of SR3 is 17.971 ± 0.002 m. The arm mode would be 1.85 mm and a curvature of -3.8 m and there would be a few percent HOM in OMC scans. **Double check values, need extracting from the plot.**

3.1.1 Measurements of the Mode Matching Between the Arm Cavities and the OMC at LLO Between 2018-2019.

The mode matching between the arm Cavities and the OMC, and the input light to the OMC has been studied at LLO. The following subsection describes measurements made at LLO between 2018-2019. Techniques for determining the mode matching between the arms and the OMC include: actuating the waist size and waist location of the beam coming out of the arm and measuring what effect this has on the mode matching to the OMC; direct beam profile measurements; the application of different DARM offsets to determine what portion of light in an OMC scan corresponds to the arm mode and what portion of the light comes from elsewhere in the interferometer (e.g. from contrast defect); and use of a beacon signal to distinguish light from the signal-carrying arm cavity light from junk light. Modelling work relies on the modematching of the PRC and the arms to be $> 99\%$ (estimated from interferometer reflectivity [15]), as this allows singlebounce and full interferometer measurements to be compared. The upper limit on mode matching from the arm to the OMC is 99% , and this was obtained from actuation on the beam parameter and the use of a beacon signal (meaning this must be arm cavity light). The entire collection of measurements find the mode matching to be between 87% and 99% . Since this is quite a large discrepancy, the measurements are briefly discussed below. *Analysis of these results is being developed (cite Carl Blairs work)?.*

Mode scans for an input power of 10W find the mismatch to be 11% [17]. The DARM offset was changed in order to determine what light was coming from the arm mode. An input power of 10W is fairly low, the design is 125W , so the thermal

state of the interferometer may be different in this measurement compared to when the interferometer is in observing mode. This method was repeated for 40W of input power[18]. In this measurement, it was found that there was 4% mode mismatch.

A direct measurement of a beam profile finds there to be A 15% mode mismatch between the OMC waist and the beam transmitted through the SRM while the interferometer was in an ITM single bounce [19][20] configuration. This measurement also shows a fair amount of astigmatism, with the waist in the x and y directions being separated by ≈ 0.5 m. The x direction data is discarded in the analysis as there was trouble orienting the beam profiler. Since there is good mode matching between the input and the arms ($> 99\%$), there should not be much difference between this beam and the beam when the interferometer is locked as long as the thermal lensing in the ITM was similar to how it would be in observation mode. This type of measurement cannot be performed while the arms are locked as they require commissioner to be working nearby the arm cavities. This analysis is in fairly close agreement with the mode scan measured in [17].

Altering the beam parameter using the ITM thermal lens actuator has shown that the mode matching between the arm cavity and the OMC can be improved by 1% or 2%[21]. Using the SRM CO2 heater increased the optical gain [21], and the amplitude of the beacon signal increased by 10%. The ITM and the SRM are separated by enough Gouy phase that actuating on them covers a sufficient amount of beam parameter phase space in determine if the mode matching was optimised. does [21] agree with [17] and [19][20]. The interferometer was in single bounce mode, but since the mode matching between the PRC and arm Cavity is high, this

is similar to a locked-arm cavity measurement. There was 25W of input power, so thermal effects may be different between this measurement and observation mode. This type of measurement was repeated using 40W of input power[22]. Assuming the ITM thermal lens was optimal, then the mode matching from the arm to the OMC was found to be optimal with no SRM heating applied. This suggests that the mode matching is $> 99\%$. A beacon signal was used to determine the mode matching from the arm to the OMC.

Modelling of the arm mode finds that a 3.5% mode matching improvement between the arm cavity and the OMC can be made using the ITM thermal lens and the SRM thermal lens [23]. This modelling is compared to the measurement made with data from [20]. It was found experimentally that a 4% improvement in the arm to OMC mode matching could be achieved for a single bounce measurement with 25W of input power. This evidence is in disagreement with [22] which shows that the mode matching is already high.

Just after the interferometer had lost lock, a series of mode scans were used to create a model which predicts 92% mode matching[24] between the beam transmitted through the SRM and the OMC. The input power was 20W and the lock lasted 1 hour and 41 minutes, so the interferometers thermal state immediately after this lock loss was the same as it would be for a typical time during observing mode when the input power was 20W. The thermal lens in the ITM was measured using the Hartmann wavefront sensors, and this was used to model the interferometer at the time of unlocking. The Hartmann data is very noisy, which could cause uncertainty in the mode matching prediction made by the model, although no error on the fit

parameters is given in [24].

The mode matching between the squeezer and the optics within HAM6 was studied [25][26]. Measuring the beam profile between the squeezer and the OMC allows for the layout in HAM6 to be modelled. This means that there is less uncertainty in where the mismatch between the arms and the OMC may come from. The mode matching between the squeezer and the OMC was measured to be 98%. The model of HAM6 that was developed from this data differs significantly from the nominal values of the optics in HAM6, a 15% error in the RoC of OM2 is required to explain the beam profile measurements. This may not be unreasonable, as accurate measurements of focal lengths for mirrors with $f = 2\text{m}$ are hard to make (ref?).

3.2 Modelling Interferometers With Mode Mismatches

Using FINESSE

Mode decomposition and mode mismatching will be described earlier in thesis. This section is specific to FINESSE.

FINESSE is software for simulating optical layouts. Optical layouts are modelled with components such as mirrors, lenses, and spaces. The way these components are connected is defined by nodes. FINESSE performs a nodal analysis to calculate the steady state amplitude of light fields within the optical layout, similar to how a circuit can be analysed. PyKat is a python wrapper for FINESSE which provides many quality of life improvements such as the ability to programatically updating and re-running simulations [27].

The light fields that are calculated in a FINESSE simulation can be measured by using detectors such as a photodiode, `pd`, or an amplitude detector, `ad`. These can be configured to detect only certain modes using the `mask` command. The `bp` detector is used for measuring the beam parameter, and the `gouy` command computes the Gouy phase accumulated along a set of spaces. These detectors were used to find the ratio of HOM to HG00 light, the beam parameter of the arm mode that is transmitted through the SRM, and the Gouy phase accumulated along the SRC.

The beam parameter at nodes in a FINESSE simulation are determined by `cav` and `gauss` commands. The `cav` command is used for defining linear and circular cavities. The beam parameter is found by solving for the cavity eigenmode. Once these nodes have had a beam parameter defined at them, the components connected to them have their beam parameter calculated based on the set of linear equations that define the component. If there are more than one `cav` commands used in a simulation, the latter `cav` command overwrites the previous one. The `gauss` command executes after the `cav` commands, and sets the beam parameter of the laser used in the simulation.

FINESSE can perform simulations using HG modes, allowing for the effects of mode mismatches to be modelled as long as the problem satisfies the paraxial approximation. A coupling between the HG modes happens whenever a beam described by q_1 enters a segment described by q_2 . This could happen if the `gauss` command used on the laser does not correspond to the eigenmode of a cavity defined by the `cav` command.

A laser beam can be modelled as a linear combination of Hermite Gaussian modes. Ideally an infinite number of these modes would be used in a calculation, however since it is not feasible to simulate an infinite number of TEM modes the maximum number of TEM modes needs to be defined in a FINESSE simulation. This is done using the `maxtem` command. FINESSE will calculate the coupling of the light amongst these modes. If an insufficient number of TEM modes is used in the simulation, the output will be nonsense as power from the higher order TEM modes that would be required to accurately model the optical layout will be ‘aliased’ into the ones defined in the simulation. When a cavity becomes unstable, FINESSE will not generate physical results since the cavity will not be mathematically describable.

The distance between components, L , in a FINESSE simulation is defined with a macroscopic distance, l , which is modelled with the `space` command, and a microscopic distance which is modelled with the tuning parameter, ϕ , belonging to components such as mirrors (`m`) and beamsplitters (`bs`). The macroscopic distance l is always an integer number of wavelengths, whereas the microscopic tuning, ϕ , determines the sub-wavelength accuracy of the length of the space. FINESSE uses the convention that 1 wavelength corresponds to 4π rad phase. It is useful to split the total length, $L = l + \phi$, in this way since the l determines things like Gouy phase, whereas ϕ determines whether a cavity is on resonance/which mode is resonant. When simulating mode mismatches, the command `phase 2` is required to give physically correct results. This sets the phase accumulated by the HG00 mode in a `space` to 0° .

3.2.1 Thin and Thick Beamsplitters

Thin beamsplitters are made with the `bs` command. Care needs to be taken when using these components since incorrect usage can yield erroneous results. To illustrate this two simulations of Michelson interferometers were run, one with a thin beam splitter and the other with a thick beamsplitter. The FINESSE code for these simulations can be found in appendix A.3.

For energy to be conserved, each reflection from a dielectric coating where the refractive index is lower on the reflecting side than on the transmitting side, a light field gains 180° of phase. In a Michelson, there is one path with 2 of the π phase change reflections, while in the other path there is only one such reflection. This means when the arms have equal lengths, destructive interference occurs at the AS port. The intensity, I , at the AS port of the Michelson as a function of the tuning of one of the arm mirrors, $\Delta\phi$, including the FINESSE convention for phase, is

$$I = I_0 \sin^2(\Delta\phi) \tag{3.1}$$

where I_0 is the input power.

The effect of using a thin beamsplitter versus a thick beamsplitter in a FINESSE simulation is shown in figure 3.1. The thin beamsplitter causes there to be constructive interference at the AS port for equal arm lengths, whereas equation 3.1 shows that it should be destructive interference. Using a thick beamsplitter fixes this ‘bug’, as the phase at the beam splitter is properly accounted for.

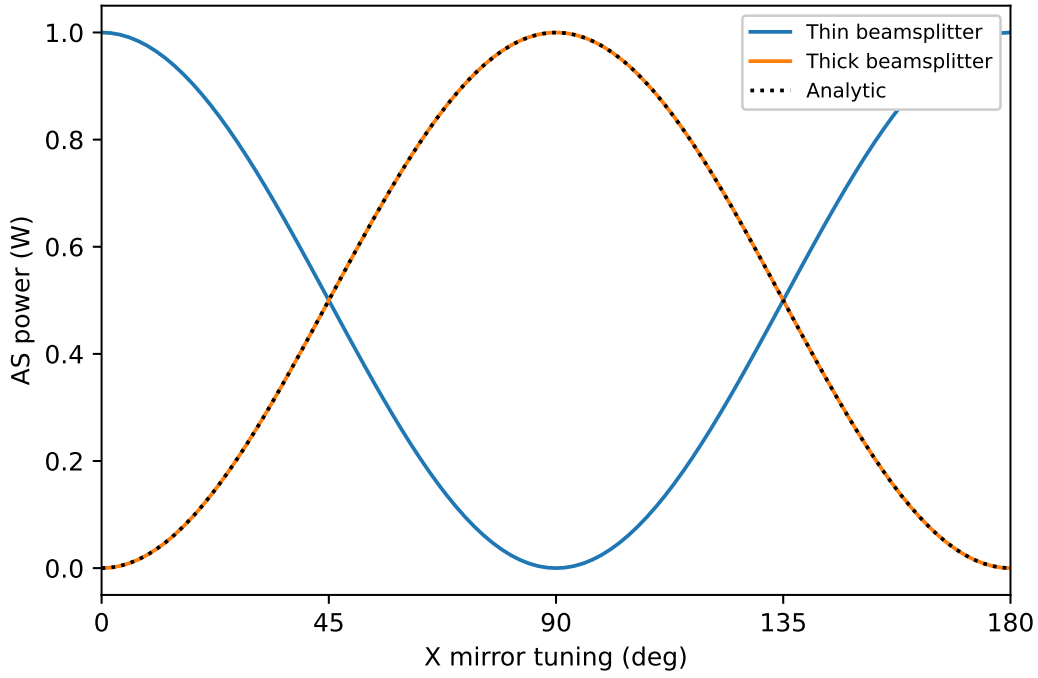


Figure 3.1: The simulation of a Michelson interferometer with thin a beamsplitter is shown in blue. The same simulation was done with a thick beam splitter, this is shown in orange. The analytic behaviour of a Michelson interferometer (equation 3.1) is shown by dashed black line. The tuning of one of the arm mirrors was changed and the power at the AS port was measured. Comparing the outputs of the simulation to equation 3.1 shows that thick beamsplitters should be used.

3.3 Modelling of Michelson Interferometer With Fabry-Perot Arm Cavities and A Signal Recycling Cavity

To investigate what effect a mode-mismatch between the SRC and the arm cavities has on the HOM content of the beam transmitted through the SRM, a ‘LIGO-like’ interferometer was simulated in FINESSE. This interferometer uses identical, design specification arm cavities and the LLO SRC. The reason for using identical, design value arm cavities is to eliminate contrast defect. The OMC, IMC, and PRC were not included in this simulation as these do not effect the beam in the SRC. To perform ITM single bounce measurements, the ETMs are misaligned resulting in the ITM having an effective power reflectivity of 1. To replicate this in FINESSE, the R for each ITM was set to 1 minus the loss of the ITM HR coating. This is valid as long as the mode-matching between the PRC and the arms is close to 100%. The interferometer was set up in DC readout mode. The tuning of the arm mirrors was set so that there was 30 mW of light emerging from the AS port with nominal spacings and optic RoC. A schematic of the simulation is shown in figure 3.2. Nominal parameters in the simulation are listed in table A.7.

Geometry errors in the SRC cause couplings between the HG00 mode and higher order modes. There are two sorts of these errors: RoC errors and spacing errors. RoC errors can be found in SR3, SR2, SRM and the off-design thermal lensing in the ITM and compensation plate, and spacing errors between each of the optics in the SRC. The largest geometry errors in the SRC come from RoC errors in SR3 and

SR2, with both errors affecting the beam in a similar way as there is not much Gouy phase between them. The thermal lensing in the ITM is assumed to be 50 km, and it would require significant deviation, in the direction of more heating, for this lens to have a large effect. The radius of curvature of SRM is well enough known that it is not the dominant cause of junk light in the SRC. The spacings between the optics in the SRC is known to a great enough precision that errors in their locations have a negligible effect. Therefore, to investigate how geometry errors effect the beam transmitted through the SRM, the RoC of SR3 and SR2 will be varied.

All simulations used the same spacings, RoC and focal lengths, apart from the focal lengths of SR3 and SR2 which was a variable for most simulations. The ETM and ITM both have 10 ppm loss. The arms are both made to be identical so there is no contrast defect, and are based off the nominal design. A 50km thermal lens is assumed in all simulations. The lengths in the SRC are ones directly obtained from the Zemax model of the LLO interferometer. To simplify the modelling, the SRC telescope is constructed using `lens` rather than `bs` with `attr` commands (although its not much more complex to use `bs`). The focal lengths in the SRC are the measured values for the LLO optics.

For simplicity, no Schnupp asymmetry was included. In a real interferometer, a Schnupp asymmetry is required to make the interferometer controllable. Since spaces are by default on resonance in FINESSE, no Schnupp asymmetry is needed to control the inner michelson length. The Rayleigh legnth of the beam at the ITM is of the order of several hundred metres which is huge in comparison to the Schnupp asymmetry, which is 8 cm. The Schnupp asymmetry will have a small effect on the

HOM content of the beam transmitted through the SRM, but since the dominant source of HOM comes from RoC errors in SR3 and SR2, the behaviour of the beam in the SRC can be largely understood without it.

The beam parameter of the input laser was defined so that there is no difference between the single bounce and the locked arm simulation. To define the beam parameter, the **gauss** command was used. In this simulation, the mode matching from the laser to the arm cavities should be perfect, the actual amount used in this simulation is 99.99999%. See appendix A.1 for details on how w_0 and z were determined for the **gauss** command used on the laser. The power of the input laser was set to 5 MW.

The **cav** command was used to set up the arm cavities, although the **gauss** command used on the laser means the **cav** command is not required (on the fly testing showed this to be the case). A **cav** command was not used on the SRC since this forces the beam parameter at the SRM to be determined by the SRC geometry. An SRC cannot be defined in FINESSE using the **cav** command as one of the end mirrors cannot be properly defined.

The detectors used to analyse the mode-matching between the nominal arm mode (which matches the laser mode) and the SRC mode were **pd** and **ad** with various **mask** commands applied to them. To detect the total amount of light, no **mask** is applied. To detect the amount of higher order light a **mask [pd] 0 0 0** command is used. This sets the sensitivity of the photodiode to the HG00 mode to 0. The mode matching is then determined as $1 - \text{HOM}/\text{total light}$. **maxtem** is set to 6. As a cross check, 2 amplitude detectors were used to detect the HG02 and HG20 modes:

```
ad ad_HOM_2_y 2 0 0 nSRMb
ad ad_HOM_2_x 0 2 0 nSRMb
```

When plotting the power of the HG20+HG02 mode, the square of the `ad` detector output needs to be taken. The `bp` detector was used to measure the beam parameter. To measure the gouy phase accumulated in the SRC by the HG00 mode, the `gouy` command was used.

Both the signal recycling and RSE modes of operation were investigated. The tuning of the SRM determines whether the interferometer is in RSE mode or in signal recycling mode. If the interferometer has arm cavities, the SRM detuning which corresponds to RSE is 90° while the detuning which gives signal recycling is 0° . For an interferometer without arm cavities, to operate in RSE mode the SRM detuning should be set to 0° , while to be in signal recycling mode the detuning of the SRM should be set to 90° .

3.3.1 Determining Tunings for DC readout

To maintain the same SRC finesse seen by the differential mode for each simulation, the interferometer set up so that it was on the dark fringe with 30 mW of light being transmitted through the SRM. This was achieved by applying a differential tuning to the ETMs for simulations with the arm cavities, and a differential tuning to the ITM for single bounce simulations. The light transmitted through the SRM for an interferometer with arm cavities as a function of differential ETM detuning is shown in figure 3.3. For an interferometer without arm cavities, the power transmitted through the SRM as a function of differential ITM detuning is shown in figure 3.4.

The tunings used to operate with 30 mW of light being transmitted through the SRM are shown in tables 3.1 and 3.2.

Mirror	SRM Tuning ($^{\circ}$)	Mirror Tuning ($^{\circ}$)
ETM _x	0	$+2.76\text{e} - 06$
ETM _y	0	$-2.76\text{e} - 06$
ETM _x	90	$+2.64\text{e} - 05$
ETM _y	90	$-2.64\text{e} - 05$

Table 3.1: The tunings used for the arm-cavity simulation to get 30mW of light at the dark port.

Mirror	SRM Tuning ($^{\circ}$)	Mirror Tuning ($^{\circ}$)
ITM _x	0	$+0.00709$
ITM _y	0	-0.00709
ITM _x	90	$+0.00074$
ITM _y	90	-0.00074

Table 3.2: The tunings used for the no-arm-cavity simulation to get 30mW of light at the dark port.

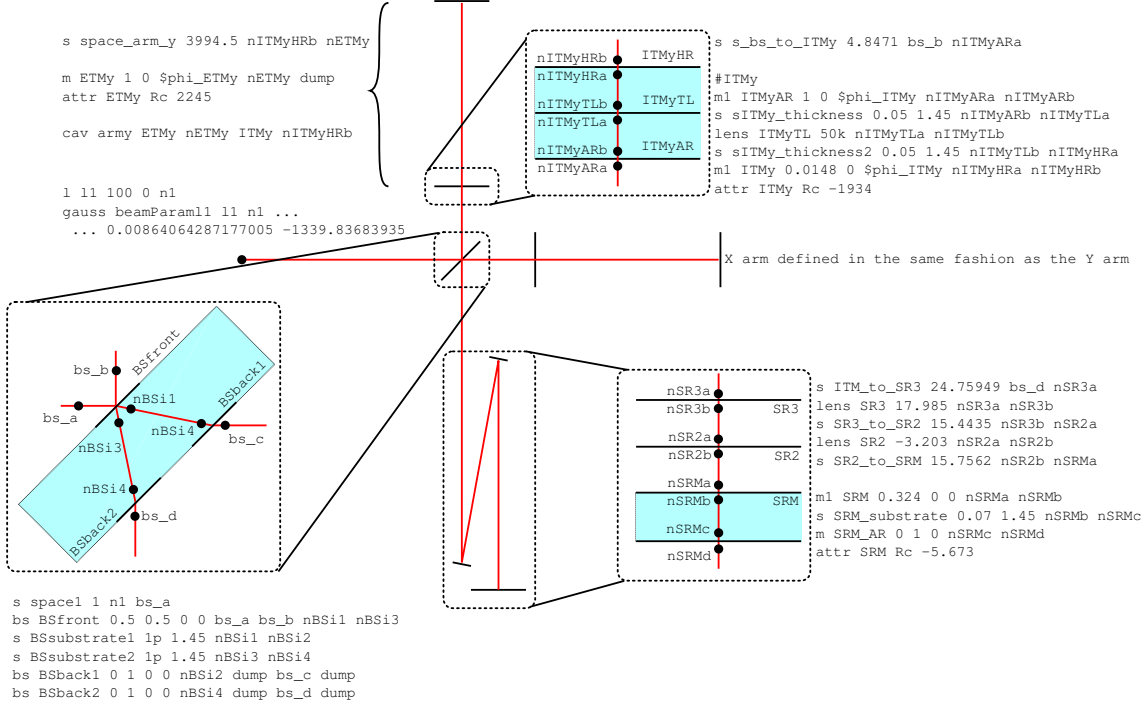


Figure 3.2: Schematic of the simulation. The dimensions of the schematic are not to scale. The laser light is shown as a red line. Black lines with red beams interacting with them corresponds to components such as mirrors and lenses. Important nodes are highlighted with black dots and the associated FINESSE commands are shown nearby. Zoom-ins provide more detail for the main beam splitter, ITMy and the SRC. The elongated curled bracket highlights the arm cavity, and the text with it shows how it is implemented in FINESSE.

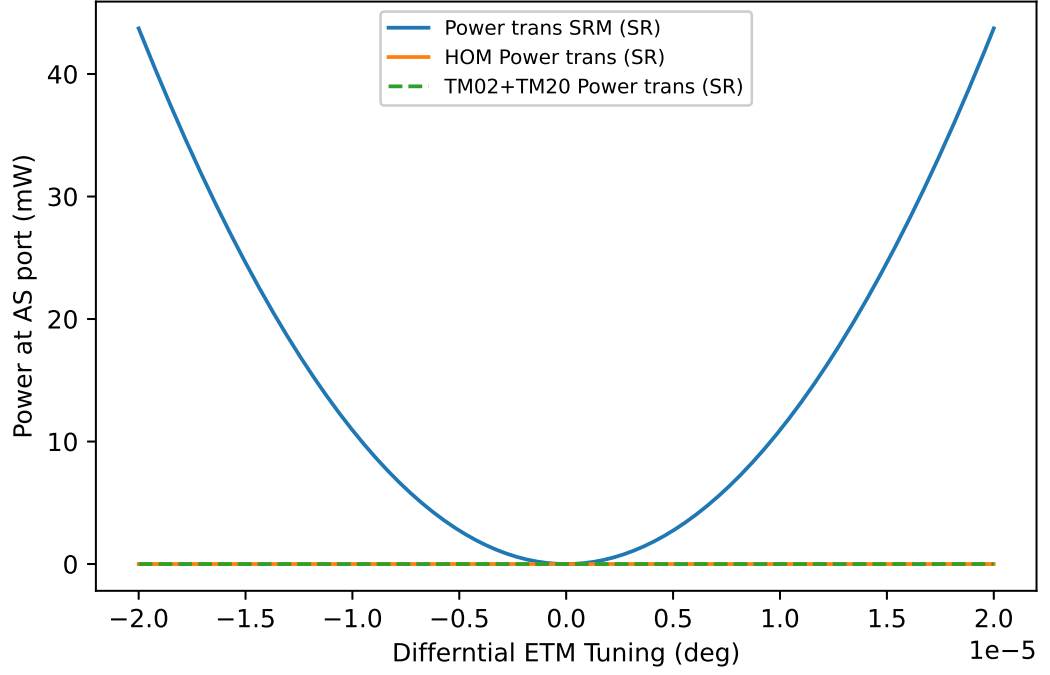


Figure 3.3: The minimum of the dark fringe is at 0° differential ETM detuning, as shown by the blue line. A slight offset causes light to be as the AS port. This result was used to determine tunings for simulating what effect focal length errors in SR3 and SR2 have on the beam transmitted through the SRM. The orange and green dashed line show the HOM content of the beam.

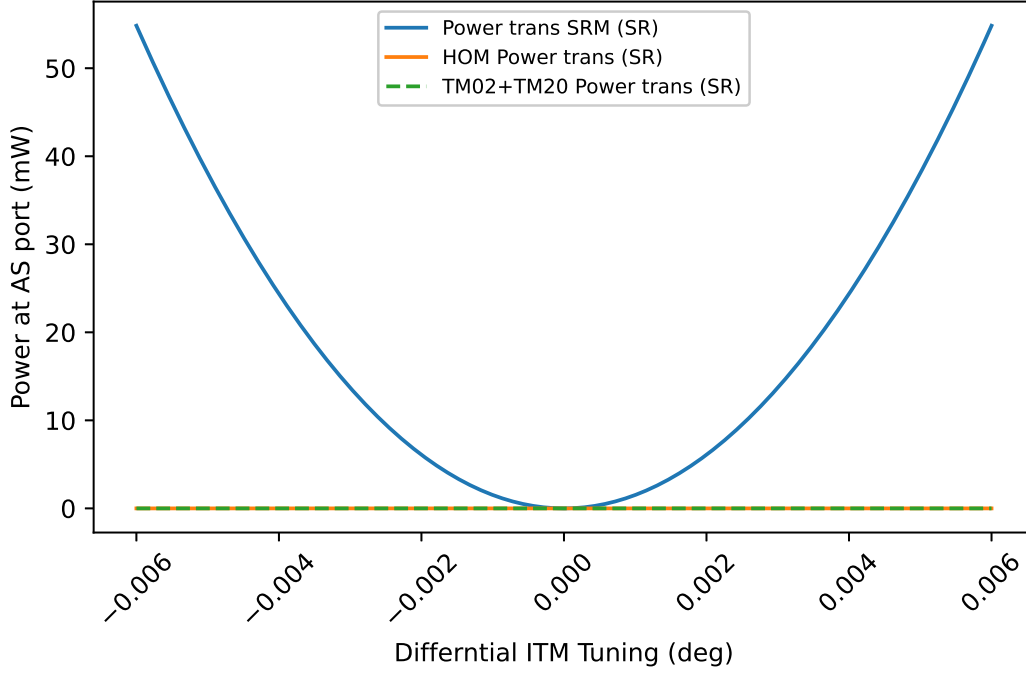


Figure 3.4: The minimum of the dark fringe is at 0° differential ITM detuning, as shown by the blue line. A slight offset causes light to be as the AS port. This result was used to determine tunings for simulating what effect focal length errors in SR3 and SR2 have on the beam transmitted through the SRM. The orange and green dashed line show the HOM content of the beam.

3.4 The Beam Parameter of the Arm Mode at the SRM

The mode defined by the arm was measured at the SRM. The beam parameter at the SRM depends on macroscopic quantities such as the distances between mirrors and the RoC of the optics, rather than microscopic tunings. Therefore, the beam parameter is not affected by the interferometer being in RSE or signal recycling mode. The beam parameter depends only on the geometry of the cavities. The focal length of SR3 and SR2 were varied around 3 standard deviations of the measured values, and the effect this had on the beam transmitted through the SRM was measured with a **bp** detector.

The effect of a focal length error in SR3 and SR2 on the width and RoC of the arm mode at the SRM is shown in figure 3.5, and 3.6. These figures show that the beam parameter depends mainly on the focal length of SR3, although SR2 has a noticeable effect. When the focal lengths of SR3 and SR2 is at the nominal values, the simulated beam has a width of 2.1 mm, a RoC of -4.0 m and the Gouy phase accumulated between the ITM and the SRM is 18.7° . There is a slight difference between the RoC found here and the design value of -3.88 m, this is because the focal lengths of the LLO SRC optics is not the same as the design values. Measurements of the single bounce beam profile[20] suggests that the beam is much larger at the SRM than the design value. This measurement also found the beam to be astigmatic, with $w_y = 2.6$ mm and $w_x = 2.4$ mm. This would correspond to a positive error in the focal lengths of SR3, as shown in figure 3.5. If the beam was this large it is unlikely that the SRC would be stable, as shown in figure 3.9.

The Gouy phase as a function of focal length error in SR3 and SR2 is shown in figure 3.7. Measurements of Gouy phase of the SRC find it to be 22.81° [28] and 21.3° [16]. These measurements of Gouy phase suggests that the beam at the SRM is smaller than the design value, and has a lower curvature. This amount of Gouy phase also corresponds to the same order of magnitude of HG00 loss that is measured and simulated (figure 3.12). The Gouy phase accumulated in the SRC is almost entirely dependent on the RoC of SR3 and SR2, positioning errors of up to 1 cm of SR3 and SR2 do not effect the Gouy phase significantly. Figure 3.7 shows that it is highly unlikely that the Gouy phase could be 22.81° in a state where the arm has greater than 90% mode matching to the SRC, as the RoC of both SR3 and SR2 would have needed to have been three standard deviations from the measured value. This could be accounted for by an much shorter thermal lens in the ITM (see figure 3.8). The measurement of 21.3° of Gouy phase is plausible with the 50 km thermal lens. The single pass thermal lens was measured to be 38.92 uD or ≈ 25.7 km [24], which corresponds to the design value and the value used in this simulation.

The focal length of SR3 and SR2 can be estimated from the Gouy phase measurement. A normal probability density function was used to model the uncertainty in the focal length of SR3 and SR2. This was combined into a 2-D normal distribution. Combinations of SR3 and SR2 that result in 21.3° of Gouy phase where found and the value of the 2-D normal distribution for those combinations was calculated. The maximum value of the resulting probability density function (which is normal) corresponds to the most likely combination of SR3 and SR2, and one standard deviation corresponds to the error. Figure 3.10 shows the that focal length of SR2 to

be -3.205 ± 0.003 m and the focal length of SR3 to be 17.971 ± 0.002 m.

The beam parameter at the SRM is determined by the focal length of SR3 and SR2. Each beam width and RoC that was computed in this simulation has a probability density associated with it (as this depends on SR3 and SR2, see figure 3.10). Therefore, the probability density function for the beam parameter of the arm mode at the SRM can be computed. Based on the measurement of Gouy phase, the beam is likely to have width 1.84 mm and a defocus of -2.80 Dioptre . This is shown in figure 3.11. This is a slight simplification as the beam parameter cannot be such that the Gouy phase of the cavity becomes too small ($< \sim 17.5^\circ$). IF the Gouy phase is too small the cavity will be unstable, so beam widths of more than 2.25 mm are very unlikely. The shape of the distribution of the arm mode beam parameter at the SRM is driven mostly by the focal length error of SR3.

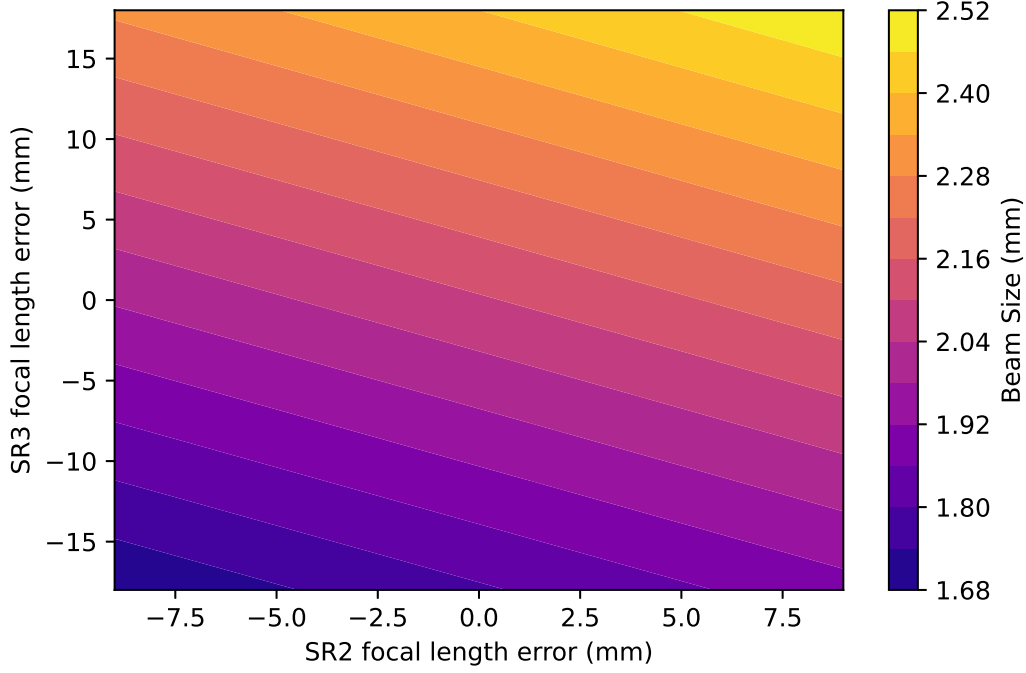


Figure 3.5: The Gaussian width of the beam that is transmitted through the SRM as a function of the error ($\pm 3\sigma$) in focal length of SR3 and SR2. The Gaussian beam width is strongly dependent on the error in SR3's focal length, whereas SR2's focal length error is less important. When the SR3 focal length error > 6 mm and SR2 error is zero the SRC cavity is unstable (see figure 3.9), so this area of the parameter space is unlikely to correspond to the beam at the SRM at LLO.

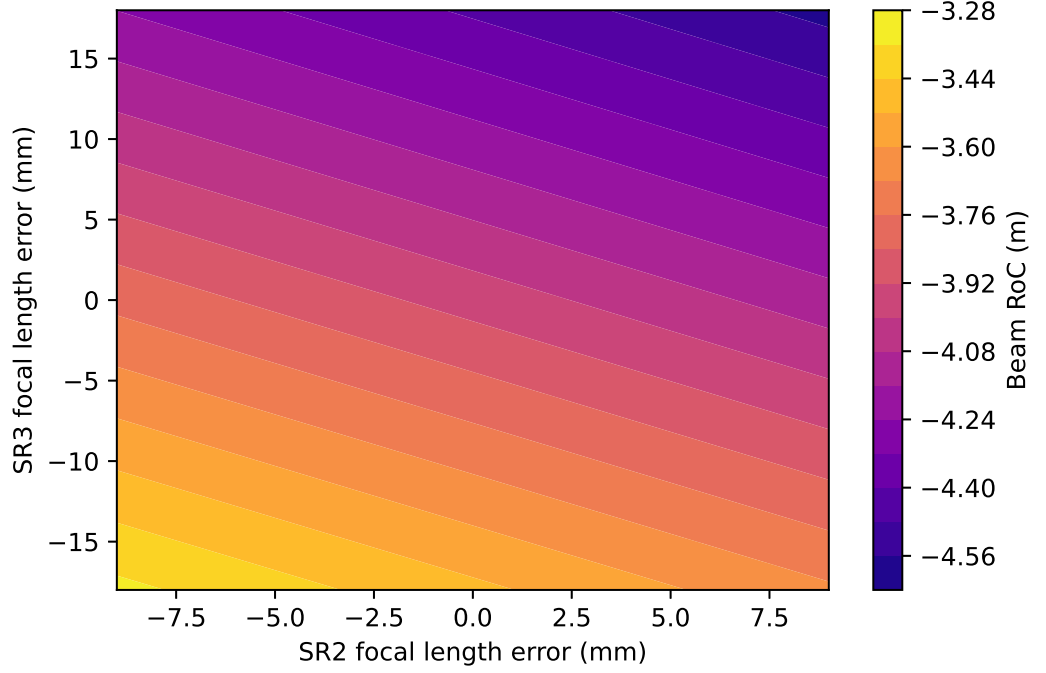


Figure 3.6: The RoC of the beam that is transmitted through the SRM as a function of the error ($\pm 3\sigma$) in focal length of SR3 and SR2. If the arm cavities and the SRC are well mode matched, then the RoC of the beam transmitted through the SRM would be $\frac{5.67\text{m}}{1.45} = 3.91\text{m}$. When the SR3 focal length error > 6 mm and SR2 error is zero the SRC cavity is unstable (see figure 3.9), so this area of the parameter space is unlikely to correspond to the beam at the SRM at LLO.

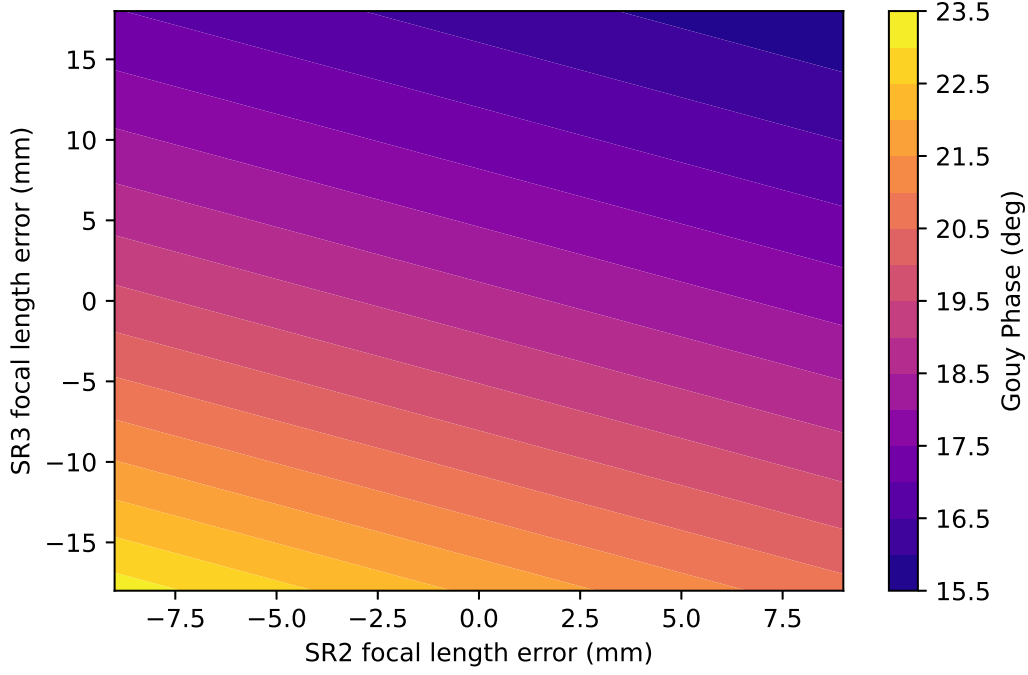


Figure 3.7: Gouy phase between the ITMs and the SRM as a function of the error ($\pm 3\sigma$) in SR3 and SR2. When the SR3 focal length error > 6 mm and SR2 error is zero the SRC cavity is unstable (see figure 3.9), so this area of the parameter space is unlikely to correspond to the beam at the SRM at LLO.

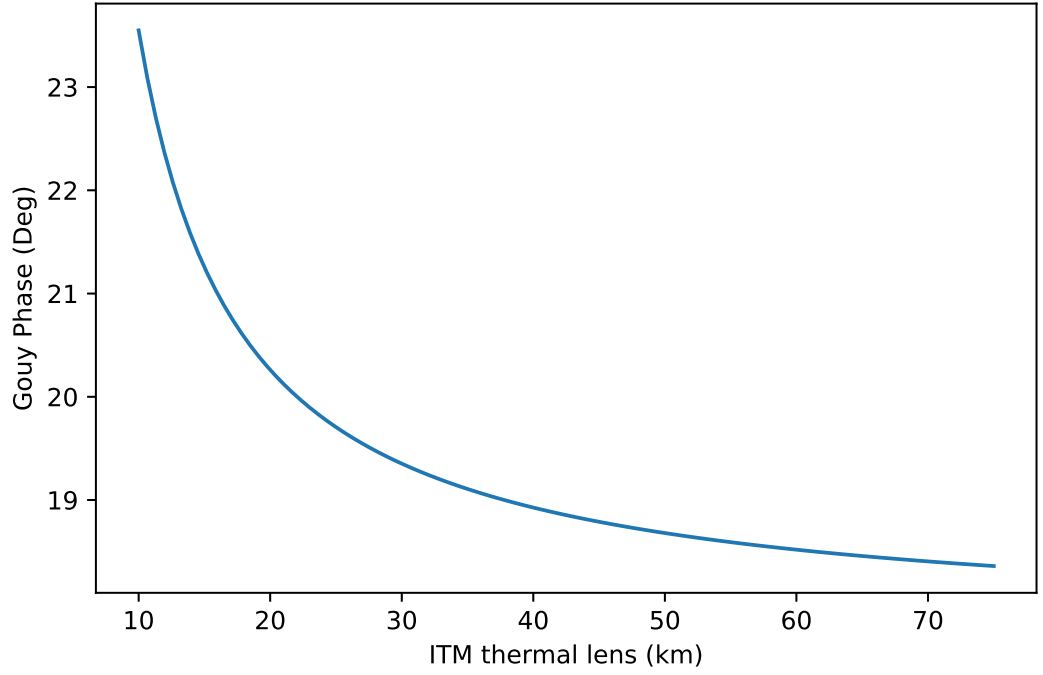


Figure 3.8: Gouy phase between the ITMs and the SRM as a function of the thermal lens in the ITMs for nominal values of SR3 and SR2. The nominal ITM thermal lens is 50 km, and this corresponds to about 19°. More heating results in a greater thermal lens, however at LIGO this is compensated for by the thermal compensation system (TCS).

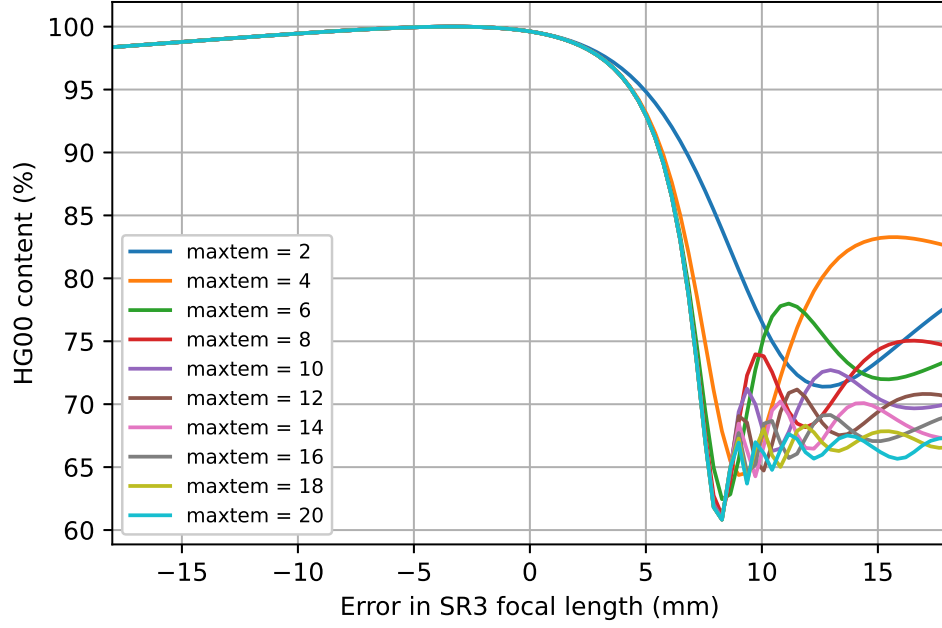


Figure 3.9: The percentage of HG00 light transmitted through the SRM as a function of SR3 focal length for the Michelson interferometer with Fabry-Perot cavities in its arms was simulated. Each simulation used a different `maxtem` command. The focal length of SR3 was changed. For focal lengths with a positive error greater than 7.5 mm, the cavity becomes unstable and FINESSE does not produce reliable results. For a stable SRC, the minimum number of TEM modes for sub 1% accuracy is six.

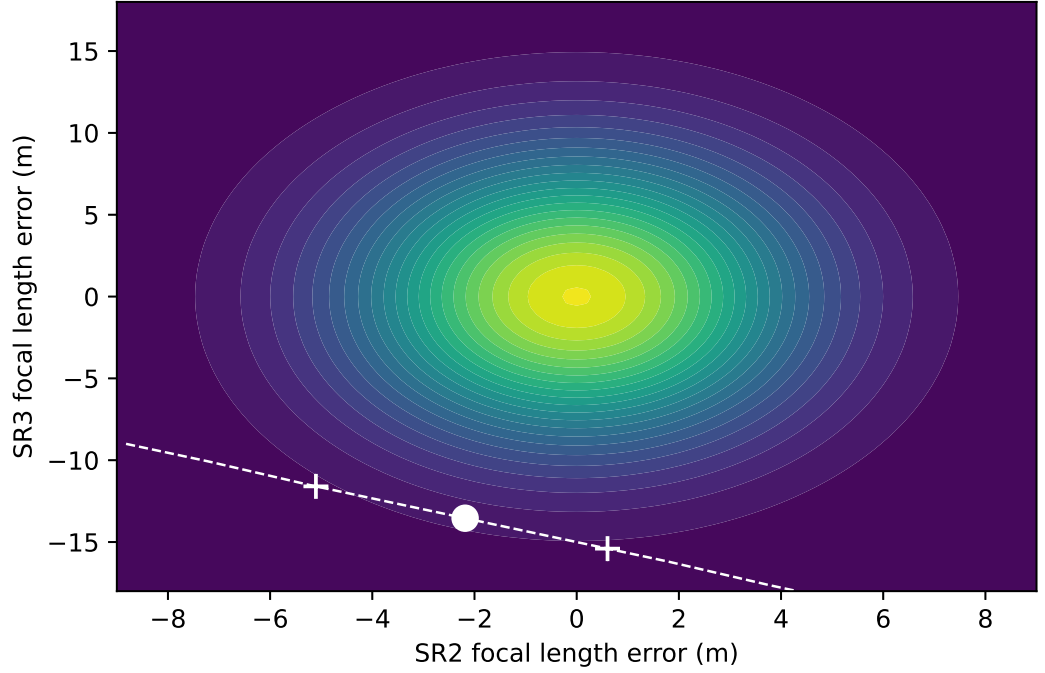


Figure 3.10: The measurement error of SR3 and SR2 follows a normal distribution. These were combined to make a 2-D normal distribution. Dark blue corresponds to the least likely combinations of SR3 and SR2, and bright yellow corresponds to the most likely. The measured Gouy phase was 21.3° [16], this is indicated by the white dashed line. The white circle marks the most likely combination of SR3 and SR2 which gives 21.3° of Gouy phase, and the white crosses show 1 standard deviation from this. The white circle and crosses indicate that the focal length of SR2 is -3.205 ± 0.003 m and the focal length of SR3 is 17.971 ± 0.002 m.

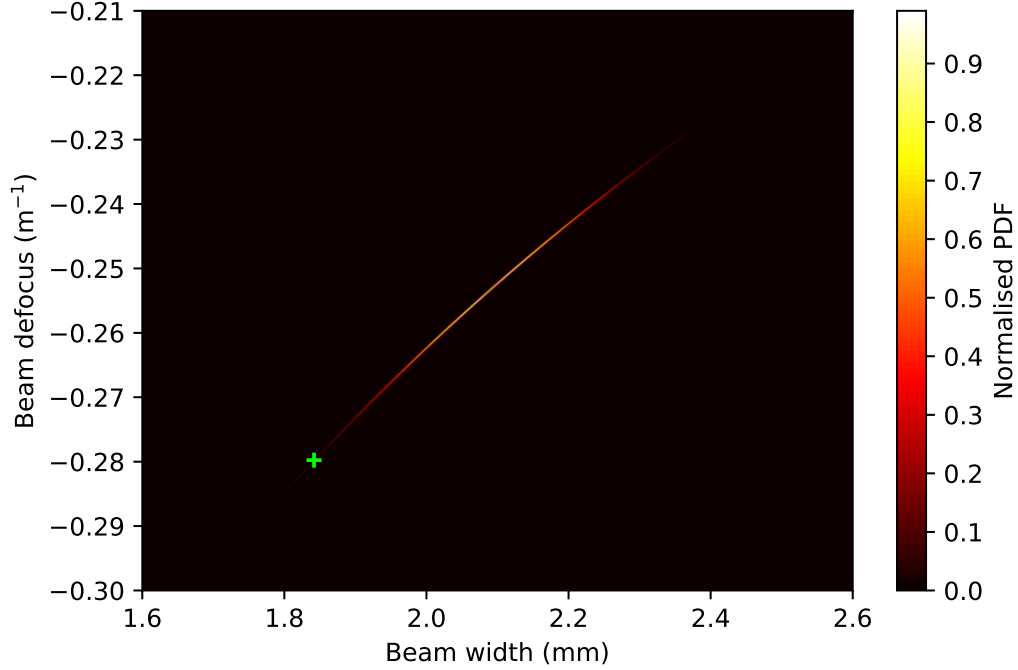


Figure 3.11: The beam parameter at the SRM was calculated for combinations of SR2 and SR3 focal lengths. The associated probability density for these combinations can be calculated (see figure 3.10), so the probability of the beam parameter having a certain width and defocus can be calculated. The probability density for each pair of beam width and defocus is indicated with the colour bar. The measured Gouy phase of 21.3° [16] is indicated with a green cross (1.84 mm, -2.80 Dioptre). This is a slightly simplified approach since the beam parameter cannot be such that the SRC is unstable. This means certain combinations of beam width and beam defocus are not possible. Beam parameters with a width $> \sim 2.25$ mm result in an unstable SRC, therefore these beams are very unlikely to be present at the SRM at LLO.

3.5 Power Transmitted Through SRM for a Mode Mismatch Between the Arm Cavity and the SRC

The HOM content of the beam as a function of the error in focal length in SR3 was investigated. As the arm cavities become more mismatched to the SRC, less HG00 mode is transmitted. The mode mismatch can be seen in figure 3.6. As the focal length of SR3 becomes further from its nominal value, the radius of curvature of the beam at the SRM stops matching the radius of curvature of the SRM. Since the SRC is a low finesse cavity, a significant amount of mismatch can be tolerated before there is a significant drop in the HG00 power transmitted through the SRM. This is shown in figure 3.12. Figures 3.12 and 3.13 show that the nominal value of the focal length SR3, 17.985 m, is slightly longer than optimal. When the SRC is on resonance, the peak in transmission is -3 mm from the nominal focal length of SR3. This is shown by the blue and green lines in figure 3.13.

The power of the HG00 light transmitted through the SRM as a function focal length error is shown in figure 3.13. In signal recycling mode, shown as blue and green in figure 3.13, the transmitted power has a peak shape. This is because an error in the focal length of SR3 moves the cavity off resonance. In RSE mode, shown as red and orange in figure 3.13, a change in focal length of SR3 does not have much effect on the amount of HG00 transmitted by the SRM since an anti-resonance is a much broader feature than a resonance.

In a single bounce measurement, the beam incident onto the ITM from outside the cavity is reflected straight back. This is equivalent to the reflectivity of the ITM

$R = 1$. When the light is resonant in the arm cavity, the ITM acts as a highly reflective mirror with some frequency dependent behaviour. This means that there is no significant difference between a single bounce measurement and a measurement with arm cavities. This is shown in figure 3.12. Higher order mode beams generated by errors in the SRC are not suppressed by the arm cavity as they are reflected by the ITM.

The ratio of the amount of light in the HG00 mode versus the total light depends on the detuning of the SRM, rather than the presence of cavities in the arm. This is shown in figure 3.12. When the light enters the arm cavities, it is reflected back with an extra π rad phase, however the HOM that does not enter the cavity does not acquire this phase. Since the HOM only interact with the SRC, the ratio of HOM to HG00 depends on the detuning of the SRM, rather than whether the interferometer is in signal recycling or RSE mode. The amount of higher order mode generated in the SRC is proportional to the HG00 power in the SRC, so the percentage of HOM beam is dependent only on the tuning of the SRM and the focal length error of SR3.

As the SRC gets closer to being unstable, more TEM modes are required to simulate the behaviour of the cavity. The effect of using more TEM modes in the simulation is shown in figure 3.9. For positive errors above 6 mm in the focal length of SR3, the cavity becomes unstable. As the error in SR3 focal length approaches this, a greater number of TEM modes is required to accurately simulate the behaviour of the interferometer. It was found empirically that six TEM modes is sufficient to simulate the interferometer while it is stable. The greyed out regions of figures 3.12 and 3.13 show where the simulation is not accurate.

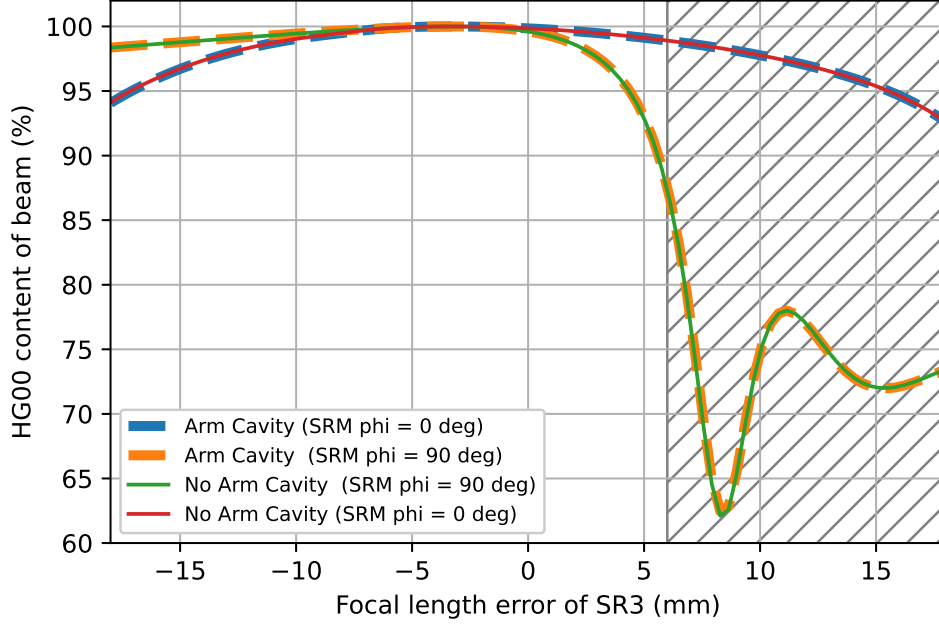


Figure 3.12: The percentage of light which is in the HG00 mode as the focal length of SR3 changes compared to the total light power does not depend on whether there are arm cavities. It is determined by the detuning of the SRM. The blue dashed line shows the behaviour of the beam with arm cavities present and with the SRC in signal recycling mode (SRM $\phi = 0^\circ$). The red line shows how the beam changes with focal length error for a single bounce measurement and with the SRC in RSE mode (SRM $\phi = 90^\circ$). The orange dashed line corresponds to the interferometer having arm cavities and being in RSE mode (SRM $\phi = 90^\circ$). The green line shows behaviour of the beam when the interferometer is configured for a single bounce measurement and is in signal recycling mode. The hatched region indicates where the SRC becomes unstable (see figure 3.9), so the behaviour of the interferometer cannot be accurately simulated for these values of focal length.

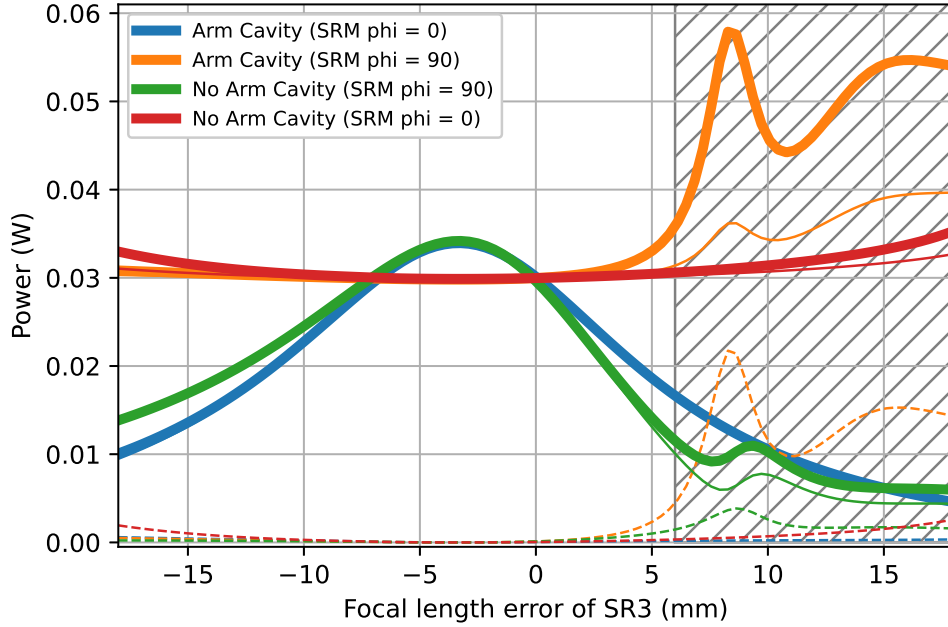


Figure 3.13: The total power transmitted through the SRM for each simulation, is shown in the thick pale lines. The thin solid lines corresponds to the power in the HG00 mode, and the thin dashed lines correspond to the power in the higher order modes. Blue lines show the behaviour when the interferometer has arm cavities and is in signal recycling mode, and the orange lines shows when it is in RSE mode. Green lines show how the power changes when the interferometer is configured for a single bounce measurement and is in signal recycling mode, and the red line shows when its in RSE mode. The hatched region indicates where the SRC is becomes unstable, and so the behaviour of the interferometer cannot be accurately simulated for these values of focal length.

3.5.1 Change of finesse of SRC

As the transmission of the SRM decreases, the finesse of the cavity increases. A FINESSE simulation was run to see the effect of increasing the power transmission of the SRM. The HG00 mode power decreases with lower SRM transmission slightly slower than the power transmitted in the HG20+02 mode. This is shown in figure 3.14.

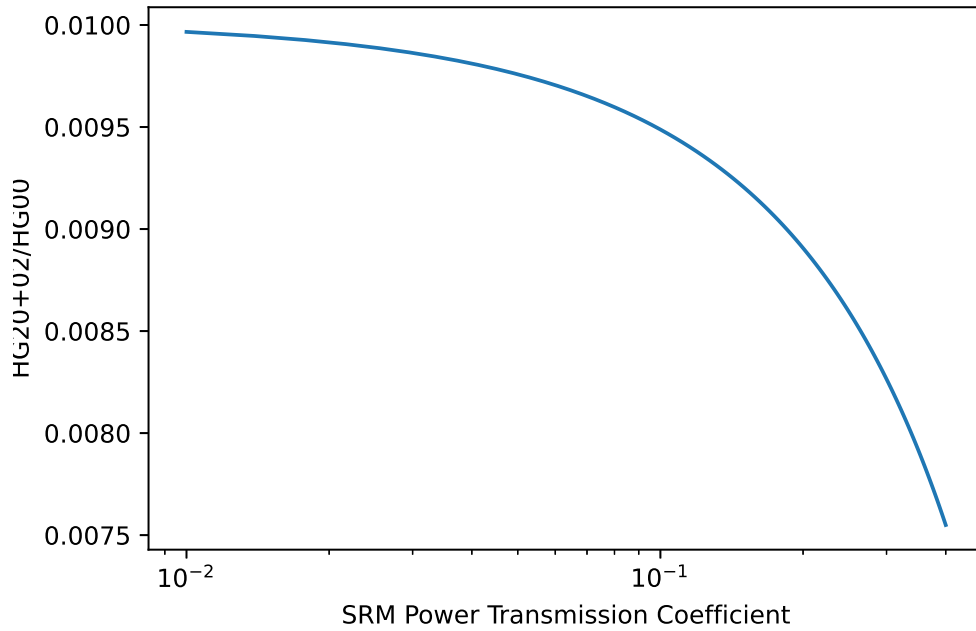


Figure 3.14: Ratio of HG20+02 to HG00 as the transmission of the SRM is increased for a mode mismatch between the arm cavity and the SRC. The interferometer is in RSE mode. A mode mismatch was introduced by setting the focal length of SR3 to be 12 mm shorter than the nominal amount.

3.6 Conclusion

Mode-mismatching between the arm cavities and the SRC results in the scattering of HG00 mode into higher order modes. Losses in HG00 usually reduces the SNR by $\sqrt{\text{Power}}$, however since there are multiple points between the squeezer and the OMC where mode-mismatches can happen, the quantum noise is enhanced by more than the usual amount for a loss. There exists a collection of measurements which show the arms are matched between 92% and 98% to the OMC.

FINESSE simulations were used to model mode-mismatches between the arm and the SRC caused by RoC errors in SR3 and SR2. The simulation used as built parameters for the SRC at the LLO site and design specification arm cavities. This allows us to study the effect of non-ideal mode-matching in a close to real life situation while reducing the complexity of the model. The interferometer was modelled in signal recycling mode and RSE mode for and locked arm cavities and single bounce configurations. It was found that there is no significant difference between single bounce and locked arm cavity configurations as there is very high levels of mode matching between the PRC and the arm cavity. In both single bounce and locked arm modes, the ITM forms a highly reflective mirror as almost all the light is reflected back. The amount of HOM is proportional to the HG00 mode in the SRC. The ratio of HOM to HG00 depends on the detuning of the SRM. As HOM do not enter the arm cavity, they cannot distinguish between the signal recycling and RSE modes of operation.

The uncertainty in the beam parameter of the arm mode at the SRM and the eigenmode of the SRC is driven by the uncertainty in SR3. This is shown in figures

3.5 and 3.6. The single pass Gouy phase of the SRC for different combinations of SR3 and SR2 was simulated, and is shown in figure 3.7. Using the measured single pass Gouy phase of the SRC, the focal length of SR2 was found to be -3.205 ± 0.003 m and the focal length of SR3 was found to be 17.971 ± 0.002 m. This is shown in figure 3.10.

Future work could involve using a heater to change the focal length of SR3. This would produce a predictable change in the focal length of SR3. This could be compared to the Gouy phase of the SRC to check that this changes the in mode matching the expected way. As the lengths between all the optics is well known, and the focal length of the ITMs is well known, the squeezer beam could be used to probe the radius of curvature of SR3 and SR2. By setting up a single bounce measurement, the incoming and outgoing beam profile could be measured and compared to a model. Having greater confidence in the beam parameter of the arm mode at the SRM would be of great benefit for the design of future upgrades to LIGO.

Appendix A

FINESSE Simulations

The following simulations were used in chapter 3. Section A.1 shows how the input beam parameter was determined so that single bounce simulations and simulations with arm cavities had the same beam parameters. Section A.2 contains tables of the spacings and properties of the optics for the LLO interferometer.

A.1 Mode Matching the Laser to the Arm Cavity

To find suitable numbers to use for the `gauss` command in the simulation of single bounce and locked arm beam measurements (see section A.3 and figure 3.2), the following simulation was run. It uses the same Y arm and beam splitter as the main simulation from section A.3, however it does not have an X arm or the SRC. Without the `cav` command, finesse will default to using a 2mm beam with its waist at $z = 0$ with respect to the node at the laser. The following beam parameters were found with a `bp` detector placed at the laser's node:

- $w_0 = 0.00864064287177005$ m

- $z = 1339.83530090614$ m

To figure out if the sign of z is correct, consider what happens if the beam parameter detector is moved 1m towards the cavity to **bs_a**:

- $w_0 = 0.00864064287177005$
- $z = 1338.83530090614$ m

z increasing suggests moving from **n1** to **bs_a** causes one to be closer to the waist. Intuitively, we know this is the case since the geometry of the cavity means the waist is within the cavity, and moving towards the cavity means you are moving towards the cavity waist. In other words, we know the beam should be getting smaller as we move from the laser to the cavity, which matches the behaviour seen here.

However, setting the beam parameter at the laser with the first set of numbers gives *terrible* mode matching. The simulation below investigates this. A photodiode which is masked from the TM00 mode

```
pd HOM bs_d
mask HOM 0 0 0
```

was placed at **bs_d** , which has the beam reflected from the arm cavity in it. The value of z for the **gauss** was varied and the power incident on **HOM** was measured. To check that enough HOM were being used, the simulation was repeated with several **maxtem** commands. The results had a quadratic fitted to them, and the minimum was taken as the optimal value. This is shown in figure A.1. The values:

- $w_0 = 0.00864064287177005$ m

- $z = -1339.83683935$ m

give more than 99.99999% mode matching, which for this work is plenty. This is close to the original values that we obtained, except for a minus sign. This is due to the way FINESSE computes the beam parameter.

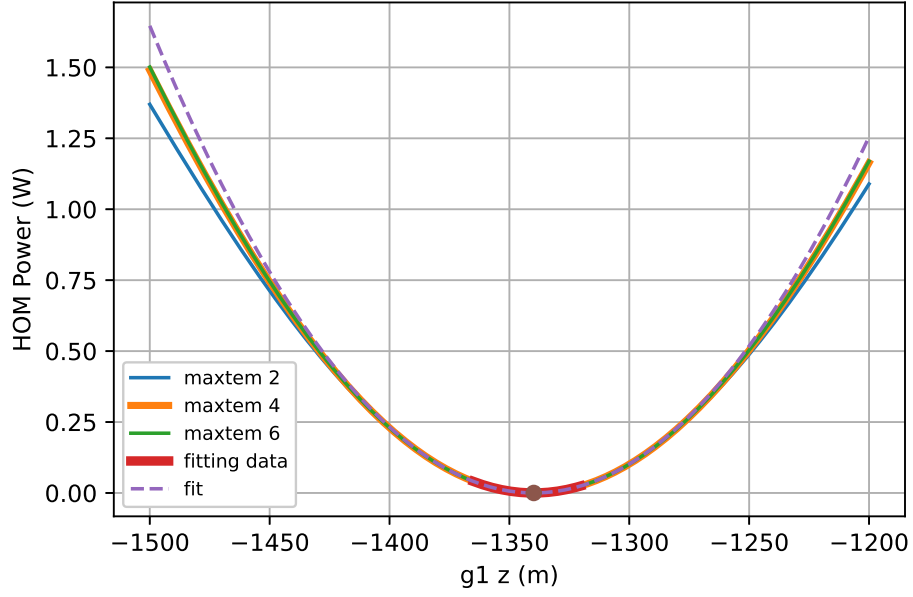


Figure A.1: Optimisation of parameters used in the `gauss` command to mode match the laser to the arms in the simulations described in chapter 3 and section A.3. The blue, green and orange lines show how the mode matching changes as the beam waist location is changed. As the mode matching becomes larger, more HOM are required, as the blue and green/orange lines diverge. A quadratic was fitted to the data highlighted in red, and the minimum value was used as the waist location parameter for the `gauss` command.

A.2 Interferometer Parameters

The simulated interferometer uses ‘design’ arms and the LLO SRC. The simulated arm cavity geometry is based on the design values to avoid contrast defects. The following subsections contain tables specifying the design and measured values for each component of the simulation.

A.2.1 Radii of Curvature of optical components

The optics in the SRC are simulated with the values from table A.2. The arm cavity optics have been simulated using the values from table A.1.

Optic Name	RoC (m)	Focal length (m)	Reference
SRM	−5.6938	−2.8469	[29]
SR2	−6.427	−3.2135	[29]
SR3	36	18	[29]
ITM	1934	967.0	[29]
ETM	2245	1122.5	[29]

Table A.1: The aLIGO design values of the RoC of the optics inside the SRC and the arm.

Optic Name	RoC (m)	Focal length (m)	Reference
SRM-02	-5.673 ± 0.002	-2.8365 ± 0.001	[30]
SR2-04	-6.406 ± 0.006	-3.203 ± 0.003	[31],[30]
SR3-01	35.97 ± 0.012	17.985 ± 0.006	[32], [30]
ITM-08	1938.44	969.22	[30]
ETM-15	2245.4	1122.7	[30]

Table A.2: The measured values of the RoC of the optics at LLO that are inside the SRC and the arm.

A.2.2 Spacings

The nominal values with a diagram can be found in [33] for the H1/L1 spacings. This information is also in [29]. The L1 Zemax parameters are found in [34]. Table A.3 contains the distances derived from the Zemax coordinates. Most spacings do not require sub mm accuracy, however SR3 to SR2 does since the Rayleigh range of this beam is very small (about 4 cm). The design value spacings are shown in table A.4. The values used in an earlier FINESSE model of LLO [4] are shown in table A.5. The thickness of each optic is found in table A.6.

Space Name	Length (m)
EMTY to ITMY	3994.4850
ITMY to CP	0.02
CP to BS HR	4.8471
BS AR to SR3	19.3659
SR3 to SR2	15.4435
SR2 to SRM	15.7562

Table A.3: Measured spacings between optics at LLO[34]

Space Name	Length (m)
EMTY to ITMY	3994.5
ITMY to CP	0.05
CP to BS HR	4.8497
BS AR to SR3	19.368
SR3 to SR2	15.4607
SR2 to SRM	15.726

Table A.4: Design value for spacings[33]

Space Name	Length (m)
EMTY to ITMY	3994.515
ITMY to CP	0.02
CP to BS HR	4.847
BS AR to SR3	19.3661
SR3 to SR2	15.4435
SR2 to SRM	15.7566

Table A.5: Values used in an earlier FINESSE model of LLO [4]

Optic	Thickness (m)
ITM	0.200
CP	0.100
BS	0.0685
Angled BS	0.1202

Table A.6: Thickness of optics between the ITM and the SRM.

A.2.3 Nominal Simulation

The nominal simulation values is shown in table A.7.

Parameter	Value	Description
SRM RoC	−5.673m	Measured RoC of SRM-02 installed at LLO
SR2 focal length	−3.203 m	Measured focal length of SR2-04 installed at LLO
SR3 focal length	17.985 m	Measured focal length of SR3-01 installed at LLO
ITM RoC	1934m	Design Value
ETM RoC	2245m	Design Value
ITM thermal lens	50 km	
ITM to ETM distance	3994.5m	Design Value
ITM to BS Y	4.8471m	As-built Value for Y arm
ITM to BS X	4.8471m	As-built Value for Y arm
BS to SR3	19.3659m	As-built Value
SR3 to SR2	15.4435m	As-built Value
SR2 to SRM	15.7562m	As-built Value
ITM T	0.0148	Power transmission coefficient
SRM T	0.324	Power transmission coefficient
ITM L	10 ppm	Scatter loss of ITM HR surface
ETM L	10 ppm	Scatter loss of ETM HR surface

Table A.7: Nominal values for the major components of the simulation. Design values were used for the arms to remove the contrast defect that would occur otherwise. The as-built values for the SRC were used to replicate the behaviour of the SRC at LLO more closely. The ITMs that are simulated have a thickness of 0.1 m and a refractive index of 1.45. The thermal lens is implemented as shown in figure 3.2 .

A.3 FINESSE Code

This section contains the FINESSE code used in chapter 3.

A.3.1 Michelson Interferometers with Thin Beamsplitter

This FINESSE code describes a Michelson interferometer where the beam splitter is implemented with a single `bs` command. This simulation will produce an incorrect result as the phase at the beamsplitter is not properly accounted for.

```
l l1 1 0 n1
s sin 1 n1 na

bs thin_bs 0.5 0.5 0 0 na nb nc nd

s Yarm 1 nb nY
m Ymirror 1 0 0 nY dump

s Xarm 1 nc nX
m Xmirror 1 0 0 nX dump

pd AS nd

xaxis Xmirror phi lin 0 180 100
```

A.3.2 Michelson Interferometers with Thick Beamsplitter

This FINESSE code describes a Michelson interferometer where the beam splitter is implemented with 3 `bs` and 2 `space` commands. This is the proper way to implement a beamsplitter in FINESSE.

```

l l1 1 0 n1
s sin 1 n1 na

bs thick_bs_1 0.5 0.5 0 0 na nb nsub1 nsub3
s bs_subX 1p nsub1 nsub2
s bs_subY 1p nsub3 nsub4
bs thick_bs_X 0 1 0 0 nsub2 dump nc dump
bs thick_bs_AS 0 1 0 0 dump nsub4 dumo nd

s Yarm 1 nb nY
m Ymirror 1 0 0 nY dump

s Xarm 1 nc nX
m Xmirror 1 0 0 nX dump

pd AS nd

xaxis Xmirror phi lin 0 180 100

```

A.3.3 Main Simulation

The following FINESSE code was used to simulate mode mismatches in the SRC.

```

l l1 5M 0 n1
gauss beamParaml1 l1 n1 0.00864064287177005 -1339.83683935

maxtem 12

s space1 1 n1 bs_a

```

```

#####

#bs beam_splitter 0.5 0.5 $phi_BS 0 bs_a bs_b bs_c bs_d

bs BSfront 0.5 0.5 0 0 bs_a bs_b nBSi1 nBSi3
s BSsubstrate1 1p 1.45 nBSi1 nBSi2
s BSsubstrate2 1p 1.45 nBSi3 nBSi4
bs BSback1 0 1 0 0 nBSi2 dump bs_c dump
bs BSback2 0 1 0 0 nBSi4 dump bs_d dump

#####

## Y ARM

s s_bs_to_ITMy 4.8471 bs_b nITMyARa

#ITMy
m1 ITMyAR 1 0 $phi_ITMy nITMyARa nITMyARb
s sITMy_thickness 0.05 1.45 nITMyARb nITMyTLa
lens ITMyTL 50k nITMyTLa nITMyTLb
s sITMy_thickness2 0.05 1.45 nITMyTLb nITMyHRa
m1 ITMy 0.0148 10u $phi_ITMy nITMyHRa nITMyHRb
attr ITMy Rc -1934

s space_arm_y 3994.5 nITMyHRb nETMy

m1 ETMy 0 10u $phi_ETMy nETMy dump
attr ETMy Rc 2245

cav army ETMy nETMy ITMy nITMyHRb

#####

```

X ARM

s s_bs_to_ITMx 4.8471 bs_c nITMxARa

#ITMx

m1 ITMxAR 1 0 \$phi_ITMx nITMxARa nITMxARb

s sITMx_thickness 0.05 1.45 nITMxARb nITMxTLa

lens ITMxTL 50k nITMxTLa nITMxTLb

s sITMx_thickness2 0.05 1.45 nITMxTLb nITMxHRa

m1 ITMx 0.0148 10u \$phi_ITMx nITMxHRa nITMxHRb

attr ITMx Rc -1934

s space_arm_x 3994.5 nITMxHRb nETMx

m1 ETMx 0 10u \$phi_ETMx nETMx dump

attr ETMx Rc 2245

cav armx ETMx nETMx ITMx nITMxHRb

#####

SRC

s bs_to_SR3 19.3659 bs_d nSR3a

lens SR3 17.985 nSR3a nSR3b

s SR3_to_SR2 15.4435 nSR3b nSR2a

```

lens SR2 -3.203 nSR2a nSR2b

s SR2_to_SRM 15.7562 nSR2b nSRMa

m1 SRM 0.324 0 0 nSRMa nSRMb
s SRM_substrate 0.07 1.45 nSRMb nSRMc
m SRM_AR 0 1 0 nSRMc nSRMd
attr SRM Rc -5.673

#cav src ITMx nITMxHRa SRM nSRMa

#####
#N.B these tunings don't represent how an IFO is kept on the dark
#fringe
const phi_ITMx 0
const phi_ITMy 0
const phi_ETMx 0
const phi_ETMy 0
const phi_BS 0
#####

beam SRM_trans nSRMd
#mask SRM_trans 0 0 0

bp SRM_trans_w0 x w0 nSRMd
bp SRM_trans_w x w nSRMd
bp SRM_trans_r x r nSRMd

bp laser_w x w0 n1
bp laser_z x z n1

```

```

beam ITMy_refl nITMyHRa
beam ITMy_refl nITMyHRa

bp ITMy_refl_w x w nITMyHRa
bp ITMx_refl_w x w nITMxHRa

pd p_in bs_a*
pd p_ref bs_a

pd p_out nSRMd

pd p_out_hOM nSRMd
mask p_out_hOM 0 0 0

ad ad_HOM_2_y 2 0 0 nSRMd
ad ad_HOM_2_x 0 2 0 nSRMd

gouy src_gouy y sITMx_thickness sITMx_thickness2 s_bs_to_ITMx
      bs_to_SR3 SR3_to_SR2 SR2_to_SRM

trace 64
phase 2

xaxis SRM_trans x lin -10 10 200

```

Appendix B

Algorithm For Finding a Three Optic Solution with 45° Gouy Phase

Separation between two of the Optics

To find a mode matching solutions between the SRM and the OMCs which have 45° phase separation for the AWC optics, an algorithm was developed. To visualise how this algorithm works see figures B.2 and B.1. Since a beam can be described by two parameters, its defocus S and width w , the modes can be shown in a 2-D space [11] (figure B.2). The action of a lens or curved mirror is to move the mode in the S direction. It is helpful to define a mode continuum along which w and/or S varies. The principle by which this algorithm works is: The continuum of modes after optic 1 and the continuum of modes just before optic 3 can be easily obtained since they must have a defined beam size, so they only vary in defocus. These two continuum are propagated to optic 2, and then its just a case of finding how to move from one continuum to an other.

The continua at optic 2 do not necessarily have to have the shape that they are sketched with in figure B.2, however if the blue-green and red-yellow continuum extend far enough they will always wrap around. It is also not guaranteed that every mode in the red-yellow can be reached from the blue-green continuum, and vice versa. Since the mapping between the blue-green and red-yellow continuum is not one to one, there is some choice in you can make. Both ‘branches’ of the continua, where the continuum turns back on itself in the w direction, do not need to be analysed at the same time, but a better solution may be found by looking at different combinations of branches. This is useful in setting a limit on the size of the continuum you search.

In our implementation of this algorithm, the continua are actually discrete, which means that when finding a mapping at optic two it is not guaranteed that a mode in the red-yellow continuum will have a corresponding mode of the same with in the blue-green continuum. Because of this granularity, if you were to plot optimal values of optic 2 and optic 3 power as a function of optic 1 power, you would see jumps in the optimal lens choices. This is especially bad around where the turning point in the continua is, since the modes defocus changes quickly while their size does not.

B.0.1 Method

This section delineates the method described in section B and can be visualised using figure B.2.

Define the target mode at the optic 3. In figure B.2, this is mode l at optic three.

Define a set of modes which share the same beam size as the target mode at optic 3 but have varying radii of curvature. In figure B.2, this is the red-yellow continuum at optic three.

Propagate the set of modes defined at optic 3 back to the optic 2. In figure B.2, this is the red-yellow continuum at optic 2.

Define the initial beam parameter and propagate this to optic 1. In figure B.2, this is mode i at optic one.

Define a set of modes that have the same beam size as the initial beam at optic 1 but differ in defocus. In figure B.2, this is the blue-green continuum at optic one.

Propagate the modes defined at the optic 1 to optic 2. In figure B.2, this is the blue-green continuum at optic two.

Calculate the difference in optical power between mode i and a mode in the blue-green continuum (e.g. mode j). In figure B.2, mode j represents just one possible mode in this continuum. Save this power.

Find a mode in the red-yellow continuum which is closest in size to mode j, call this mode k. In figure B.2, there is another possible choice on the red-yellow continua, however only one of them needs to be considered.

Calculate the difference in optical power between mode j and mode k at optic 2. Save this power.

Calculate the difference in optical power mode k and the target mode at optic 3. Save this power.

Calculate the Gouy phase separation of optic 2 and optic 3.

Repeat for every mode in the blue-green continuum, or until a gouy phase separation of 45degree has been found.

The number of modes that are considered and the range of modes in the continua should be tailored for the optical layout that is being considered. Our problem was with metre scale spacings between optics, and with metre scale Rayleigh lengths. It was found that continua with a range of the order 1-10 dioptres were suitable. This does not mean that an adaptive optic would need this actuation range, rather it allowed for the static radii of curvature for each optic that gives optimal Gouy phase separation between the active mode matching elements to be found. Using this algorithm, we were able to rapidly develop many possible HAM6 layouts. This allows for more time to consider various engineering factors (suspension size and the like) and how to adapt the mirror strengths accordingly to accommodate for small changes in a tentative design.

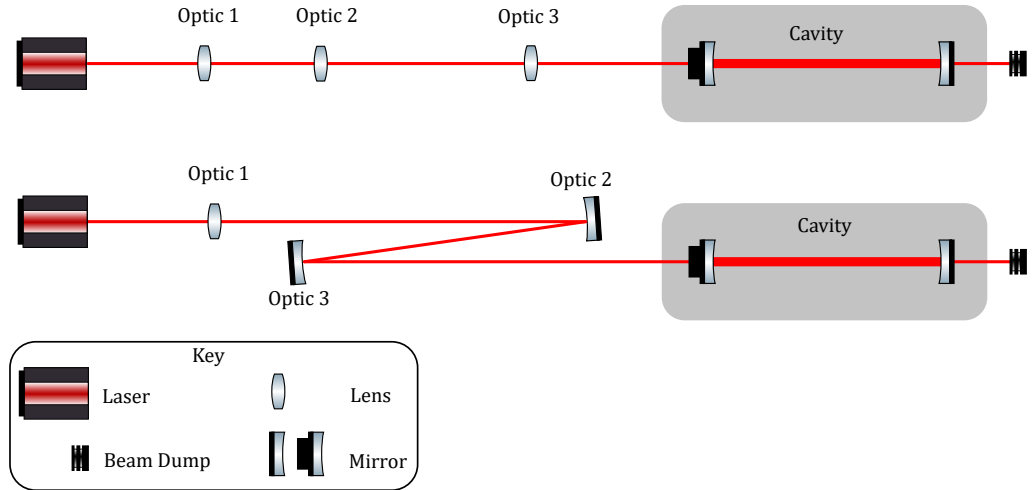


Figure B.1: Top layout: Three lenses used to achieve mode matching to the cavity. These are shown as positive lenses, however negative lenses can also be used when designing mode matching systems. Bottom Layout: A lens and two curved mirrors to achieve mode matching to a cavity. The optics 1,2 and 3 correspond to the ones shown in figure B.2. The initial and target modes are defined in these examples by the laser mode and the cavity mode.

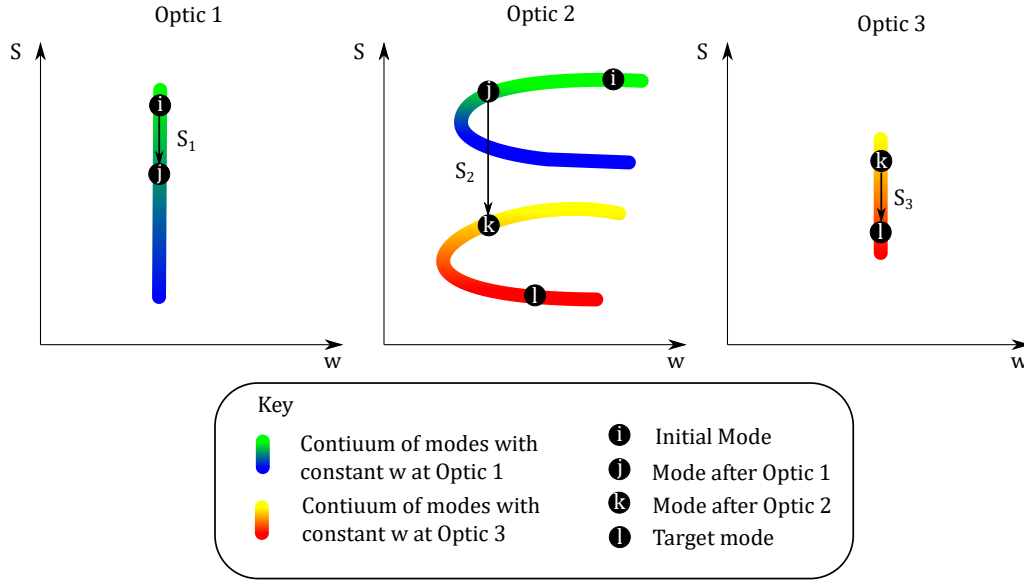


Figure B.2: Sketch of the WS phase space representations of a Gaussian beams at the three positions along the optical axis where a mirror or a lens could be. Examples of such layouts are shown in figure B.1. The optical power of the first optic is given by the difference in the defocus between the initial mode (i) and the second mode (j). The power of the second optic is found by finding the difference in defocus from the blue-green continuum to the red-yellow continuum. There are 2 optical powers that will move the mode j to the red-yellow continuum, and two possible modes in the blue-green continuum that can from which k can be reached, giving 4 possible similar solutions. The algorithm can be set up so that it is only one way of moving between the blue-green and red-yellow continua by limiting their extent. The power of the third optic is found from the difference between the mode k and mode l at optic 3.

Appendix C

Mode Matching Between 2 Astigmatic Beams

C.1 Derivation of Mode Matching TM00 Beams

The [check this naming](#) electric field, E , of a TM00 Gaussian beam can be written as

$$E(x, y, z) = E_x(x, z)E_y(y, z) \quad (\text{C.1})$$

where

$$E_x(x, z) = \frac{1}{\sqrt{q_x(z)}} \exp\left(-ik \frac{x^2}{2q_x(z)}\right), \quad (\text{C.2})$$

$$E_y(y, z) = \frac{1}{\sqrt{q_y(z)}} \exp\left(-ik \frac{y^2}{2q_y(z)}\right). \quad (\text{C.3})$$

For a beam without astigmatism, $x^2 + y^2 = r^2$ and $q_x = q_y = q$ so:

$$E(r, z) = \frac{1}{q(z)} \exp\left(-ik \frac{r^2}{2q(z)}\right). \quad (\text{C.4})$$

The mode matching, $\mathcal{M}$, of two beams can be expressed by overlap of the two fields

$$\mathcal{M} = \frac{|\int_A E_1^* E_2 dA|}{\int_A |E_1|^2 dA \int_A |E_2|^2 dA}. \quad (\text{C.5})$$

The numerator of equation C.5 for TM00 beams can be written as:

$$\left| \int_A E_1^* E_2 dA \right| = \int_A E_1^* E_2 dA \int_A E_1 E_2^* dA \quad (\text{C.6})$$

$$= \int_{-\infty}^{\infty} E_{x1}^* E_{x2} dx \int_{-\infty}^{\infty} E_{y1}^* E_{y2} dy \int_{-\infty}^{\infty} E_{x1} E_{x2}^* dx \int_{-\infty}^{\infty} E_{y1} E_{y2}^* dy. \quad (\text{C.7})$$

Since the calculation is the same for the x and y coordinates, we only need to show the solution for the x . The x terms of equation C.7 can be written as:

$$\int_{-\infty}^{\infty} E_{x1}^* E_{x2} dx \int_{-\infty}^{\infty} E_{x1} E_{x2}^* dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{q_{x1}^*}} \exp\left(\frac{ikx^2}{2q_{x1}^*}\right) \frac{1}{\sqrt{q_{x2}}} \exp\left(\frac{-ikx^2}{2q_{x2}}\right) dx \quad (\text{C.8})$$

$$\times \int_{-\infty}^{\infty} \frac{1}{\sqrt{q_{x1}}} \exp\left(\frac{-ikx^2}{2q_{x1}}\right) \frac{1}{\sqrt{q_{x2}^*}} \exp\left(\frac{ikx^2}{2q_{x2}^*}\right) dx \quad (\text{C.9})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2}{2q_{x1}^*}\right) \exp\left(\frac{-ikx^2}{2q_{x2}^*}\right) dx \times c.c. \right) \quad (\text{C.10})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2}{2q_{x1}^*} - \frac{-ikx^2}{2q_{x2}^*}\right) dx \times c.c. \right) \quad (\text{C.11})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2 (q_{x2} - qx1^*)}{2q_{x1}^* q_{x2}}\right) dx \times c.c. \right). \quad (\text{C.12})$$

Using the identity

$$\mathcal{I} = \int_{-\infty}^{\infty} \exp(iax^2) dx = \frac{i\pi}{a} \quad (\text{C.13})$$

and labelling

$$a = \frac{k(q_{x2} - q_{x1}^*)}{2q_{x1}^* q_{x2}} \quad (\text{C.14})$$

allows equation C.12 to be simplified to

$$\frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2(q_{x2} - q_{x1}^*)}{2q_{x1}^* q_{x2}}\right) dx \times c.c. \right) = \frac{1}{|q_{x1}| |q_{x2}|} \sqrt{\frac{i\pi(2q_{x1}^* q_{x2})}{k(q_{x2} - q_{x1}^*)}} \sqrt{\frac{-i\pi(2q_{x1} q_{x2}^*)}{k(q_{x2}^* - q_{x1})}} \quad (\text{C.15})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \sqrt{\frac{4\pi |q_{x1}|^2 |q_{x2}|^2}{k^2 |q_{x2} - q_{x1}^*|^2}} \quad (\text{C.16})$$

$$= \frac{2\pi}{k} \frac{1}{|q_{x2} - q_{x1}^*|}. \quad (\text{C.17})$$

This means that the numerator of equation C.5 is

$$\frac{4\pi^2}{k^2} \frac{1}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|}. \quad (\text{C.18})$$

Since E_x and E_y are orthogonal, $|E|^2 = |E_x|^2 |E_y|^2$, an expression for the denominator of equation C.5 can be found by:

$$\int_A |E|^2 dA = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |E_x|^2 |E_y|^2 dx dy \quad (\text{C.19})$$

$$= \int_{-\infty}^{\infty} \frac{1}{|q_x|} \exp\left(\frac{ikx^2(q_x - q_x^*)}{2q_x q_x^*}\right) \times y \text{ term.} \quad (\text{C.20})$$

Using a similar approach as in the calculation of the numerator terms:

$$\int_A |E|^2 dA = \frac{1}{|q_x|} \sqrt{\frac{i2\pi |q_x|^2}{k(q_x - q_x^*)}} \frac{1}{|q_y|} \sqrt{\frac{i2\pi |q_y|^2}{k(q_y - q_y^*)}} \quad (\text{C.21})$$

$$= -\frac{2\pi}{k} \frac{1}{\sqrt{(q_x - q_x^*)(q_y - q_y^*)}} \quad (\text{C.22})$$

we get the mode matching $\mathcal{M}$ to be

$$\mathcal{M} = \frac{\frac{4\pi}{k^2}}{-\frac{2\pi}{k} \times -\frac{2\pi}{k}} \frac{\sqrt{(q_{x1} - q_{x1}^*)(q_{y1} - q_{y1}^*)} \sqrt{(q_{x2} - q_{x2}^*)(q_{y2} - q_{y2}^*)}}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|} \quad (\text{C.23})$$

$$= \frac{\sqrt{(q_{x1} - q_{x1}^*)(q_{y1} - q_{y1}^*)(q_{x2} - q_{x2}^*)(q_{y2} - q_{y2}^*)}}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|}. \quad (\text{C.24})$$

C.2 Effect of Error in the Radius of Curvature of a Mirror on the defocus of a Beam

The error in the optical power of a mirror due to an error in the radius of curvature of is

$$S = \frac{2}{R_m} \quad (\text{C.25})$$

$$dS = \frac{-2}{R_m^2} dR. \quad (\text{C.26})$$

The error in the beams defocus is the same as the error in the radius of curvature of the the mirror. The effect this has on the beam parameter q can be found by

defining a contour of modes distance dq from q i.e. $q + |dq| e^{i\theta}$, where dq is

$$q = \frac{q_i}{Sq_i + 1} \quad (\text{C.27})$$

$$dq = \frac{q_i^2}{(-Sq_i + 1)^2} dS. \quad (\text{C.28})$$

This gives you a family of modes which will contain the modes due to an error in the radius of curvature of the mirror, but since q is a complex number the contour you need to define to encapsulate the concept of an error in q also contains modes which are have different beam sizes as well as different defocuses. To do the calculation this way is cumbersome, it is much more straight forward to compute the effect of having a mirror ROC error by evaluating at every mirror

$$\begin{pmatrix} 1 & 0 \\ \pm dS & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -S & 1 \end{pmatrix} \begin{pmatrix} q \\ 1 \end{pmatrix} = \begin{pmatrix} q \\ (S \pm dS)q + 1 \end{pmatrix} \quad (\text{C.29})$$

and then divide the top and bottom entries of the $\begin{pmatrix} q \\ (S \pm dS)q + 1 \end{pmatrix}$ vector.

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